

To become or not to become French: Conscription, citizenship, and labor market integration*

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Abstract

We examine how changing the costs of acquiring citizenship translates into naturalization decisions for second-generation immigrants, and the effect of naturalization on their labor market outcomes. We exploit the abolition of mandatory military service in France as an exogenous reduction in the cost of citizenship for men. In line with the predictions of our theoretical framework, we find that the reform induced a jump in male naturalization rates, entirely driven by European Union citizens. Using a Synthetic Difference-in-Differences, we show that the probability of employment for EU males consequently increased by 1.7 percentage points, through a reduction in inactivity rather than unemployment. We provide suggestive evidence that this effect is mainly driven by an increase in public-sector employment and a reduction in self-employment.

JEL classification: J15, J21, J24, J61

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1 Introduction

The continuous growth of migration flows raises crucial questions on the most adapted integration and regulation policies to implement. Naturalization, which represents the final legal step in the integration process for migrants, has consistently been a focal point of this debate. On the one hand, naturalization may be seen as a policy tool to boost migrants' integration. On the other hand, it is considered a reward for a successful integration. The latter standpoint currently guides the dominant policy-making approach. Host countries impose substantial costs on citizenship acquisition, intending to screen immigrants.

However, there is limited causal evidence on how migrants respond to changes in the cost of naturalization, and on the impact of naturalization on the labor market integration of second-generation migrants. This is due to three main challenges. First, naturalization take-up is an endogenous decision that raises concerns of selection bias. Second, exogenous shocks in existing studies often impact cohorts that are still too young for their labor market outcomes to be studied. Third, studies on citizenship acquisition for second-generation immigrants, which in most cases rely on reforms impacting individuals at young ages, may not fully be able to disentangle its effects on education from its direct labor market effects.

In this paper, we overcome these challenges by relying on two key aspects of the French context. First, individuals born in France to foreign parents face almost no costs in acquiring French citizenship at the age of 18. Second, during the 1990s, compulsory conscription for male citizens made naturalization a costly choice for foreign men. Altogether, these two features induced a salient trade-off for second-generation men at the age of 18 between renouncing French citizenship and doing military service. In this context, we leverage the abolition of compulsory conscription in 1997 for men born after 1978 as an exogenous shock in the costs of acquiring French citizenship for children of immigrants. We exploit administrative and survey data to explore how this reform affected the take-up of citizenship, and the potential repercussions on labor market outcomes.

Our results show that the abolition of compulsory conscription induced a sharp increase in naturalization rates for males relative to females. This effect is entirely driven by European Union citizens, the group of second-generation immigrants for whom acquiring French citizenship should matter the least. Yet, we find that this surge in naturalization induced a significant 1.7 percentage point increase in employment for this group. We show that these results can be explained by an expanded access to public sector jobs and a departure from self-employment. We rule out the possibility that these are driven by a direct impact of military service. These results extend the existing literature by providing quasi-experimental evidence on how changes in the costs of naturalization translate into naturalization decisions, and on the causal effect of naturalization on the labor mar-

ket outcomes of second-generation immigrants, as well as exploring potential underlying mechanisms.

To rationalize the decision to naturalize, we introduce a theoretical framework in which individuals take up citizenship as long as the benefits exceed the costs. We consider a cost function that is decreasing with skills, which is typically the case of policies such as language tests or financial requirements. The model predicts that if benefits are homogeneous, such costs would screen the top of the skill distribution. However, if benefits are heterogeneous, such costs may screen the bottom of the skill distribution by excluding low-benefit high-skilled individuals. This scenario applies to settings where groups that are the most discriminated against on the labor market are also the lower educated.

In the French context, we expect European Union (EU) citizens to benefit the least from citizenship. Indeed, unlike individuals from other birth nationalities, they can freely work and reside in France. Thus, they are the least likely to take up citizenship under compulsory conscription, and the most likely to react to its abolition. On the contrary, we expect individuals from nationalities that are typically discriminated against in the labor market, such as African nationalities, to benefit the most from citizenship. Costs are likely to be heterogeneous as well because low-educated conscripts were typically assigned more strenuous positions (Maurin and Xenogiani, 2007). As a result, we expect the abolition of compulsory conscription to have a larger impact on take-up at the bottom of the education distribution within groups of birth nationalities.

To test these hypotheses empirically, we exploit the fact that women were exempt from compulsory conscription and therefore unaffected by its abolition. Using a Difference-in-Differences approach, we compare the naturalization rate of foreign men and women born in France before and after December 31st, 1978. We find that at the abolition of compulsory conscription, the naturalization rate of males increased from 68.5% to 78.9%, while the rate for females remained stable at around 84%. This suggests that almost a quarter of the missing citizenship take-up among young males was due to compulsory conscription and that its abolition halved the gap with women.

Consistent with our theoretical framework, results show that this effect is entirely driven by European Union citizens at birth, for whom the benefits of acquiring French citizenship are lower. Within this group, the abolition of compulsory conscription increased male naturalization rates by 11.9 percentage points. No significant effect is observed for other birth nationalities, for which the cost of military service is therefore not binding. Among EU citizens, we find that the increase in naturalization rates is more than 50% larger for low-educated individuals compared to high-educated individuals, supporting the hypothesis that the cost of military service is lower for the latter.

We then take advantage of the fact that only EU males experienced a jump in naturalization to study its effect on their labor market outcomes. Specifically, we exploit every unaffected group in a Synthetic Difference-in-Differences approach to best capture how

the outcomes of EU males would have evolved absent the abolition of compulsory conscription. The synthetic control group closely mirrors the trend in the employment rate of EU males until the reform, after which the employment rate in the treated group diverged from its path with a 1.7 percentage point increase. Given that this effect is driven by 11.9% of EU males who reacted to the reform, it corresponds to a 14.5 percentage point increase among compliers. We show that this positive effect on employment is primarily attributable to a decrease in inactivity rather than in unemployment.

We explore two potential mechanisms explaining these results. First, we observe a significant increase in the probability of being employed in the public sector. Second, we find a decrease in self-employment for EU males relative to the control groups, in line with the idea that citizenship acquisition expands the set of labor market opportunities for naturalized individuals.

We conduct sensitivity checks showing that our results are robust to the set of control groups considered in our Synthetic Difference-in-Differences setting, to anticipation effects, to general equilibrium effects, to differential attrition, and to the relative length of military service in the origin country nationality. In addition, we address the concern that compulsory conscription might directly impact educational and labor market outcomes. Conscription has been shown to have either no impact or a positive impact on educational outcomes, in line with draft avoidance behavior, and on labor market outcomes in the French context ([Maurin and Xenogiani, 2007](#); [Mouganie, 2020](#)). We provide evidence that second-generation immigrants' education levels were not impacted by compulsory conscription. We also rule out a potential direct effect on labor market outcomes given the absence of labor market effects for the birth nationality groups which only experienced the abolition of military service but no changes in naturalization take-up. In addition, [Mouganie \(2020\)](#) documents that in France, military service had either no effect or positive effects on labor market outcomes. Given these factors, our study may, if anything, underestimate the actual labor market impact of naturalization.

This paper relates and contributes to three different strands of the literature. First, it sheds light on the effects of acquiring citizenship on the labor market integration of second-generation migrants. The related literature has largely focused on first-generation, establishing a positive correlation between naturalization and labor market outcomes, starting with the work of [Chiswick \(1978\)](#). An emerging strand of this literature has explored the causal link between the two, for first-generation immigrants and refugees ([Gathmann and Keller, 2018](#); [Hainmueller et al., 2019](#); [Govind, 2021](#); [Fasani et al., 2023](#); [Hainmueller et al., 2023](#)). We contribute to this literature by studying the labor market integration of second-generation immigrants, touching upon the literature on birthright citizenship which has so far focused on educational outcomes ([Felfe et al., 2020, 2021](#); [Dahl et al., 2022](#)). Our findings demonstrate that even populations who might have less to gain from naturalization, here second-generation EU citizens, experience improved economic

integration from naturalization.

Second, this paper contributes to the literature on citizenship take-up. Various studies have explored the association between the propensity to naturalize and individuals' or origin countries' characteristics such as age at migration, gender, educational attainment, and political conditions in the home country (Yang 1994; Chiswick and Miller 2009; Fougère and Safi 2009. See Gathmann and Garbers (2023) for a detailed review of the literature). In addition, citizenship acquisition costs such as civic knowledge requirements, naturalization fees, and multiple citizenship restrictions, have been shown to directly affect take-up, especially that of low-educated individuals and EU citizens (Yasenov et al., 2019; Peters and Vink, 2023; Vink et al., 2021). This paper contributes to the existing literature by formalizing citizenship take-up with a cost-benefit theoretical framework. We discuss the unintended implications of increasing naturalization costs in the face of heterogeneous benefits, echoing recent evidence of potential backlash of integration policies (Fouka, 2020; Dahl et al., 2022; Arendt et al., forthcoming).

Third, we contribute to the literature on the effects of military service. Existing research has mainly focused on the impact of conscription on citizens' outcomes such as education, employment, earnings, political behavior, and crime (e.g., Angrist 1990; Bauer et al. 2012; Card and Cardoso 2012; Hubers and Webbink 2015; Hjalmarsson and Lindquist 2019; Savcic et al. 2023, and more specifically on France: Maurin and Xenogiani 2007; Fize and Louis-Sidois 2020; Mouganie 2020). To the best of our knowledge, we are the first to investigate the effects of military service on non-citizens, and more specifically its implications for their naturalization decisions.

The remainder of this paper is structured as follows. Section 2 provides background information on military service in France, the abolition of compulsory conscription, and the relevant naturalization process. Section 3 presents the data sources. Section 4 outlines our theoretical and empirical framework for studying citizenship take-up, and estimates the effect of the abolition of compulsory conscription on naturalization rates. Section 5 describes the empirical approach used to study the impact of naturalization on labor market outcomes, presents the results, and explores potential underlying mechanisms. Section 6 provides various robustness analyses, and Section 7 concludes.

2 Context of the reform

2.1 Military service

France has had an organized military service system since the end of the 18th century. All French men were obliged to enroll in military service at the age of majority, with a possibility to postpone conscription up to the age of 22. As of 1971, women could also participate in military service on a voluntary basis.

In the case of dual citizenship, the country in which a person was required to fulfill his military service obligation could vary depending on the existence or absence of bilateral agreements between France and the country of origin. Thirteen European countries signed on May 6th, 1963, and progressively applied a convention that stated that individuals with dual nationalities from these countries were required to fulfill their military obligations in the country of usual residence.¹ Similar conventions were also ratified bilaterally with Algeria, Israel, Switzerland, and Tunisia.²

With a decline in military needs, the length of conscription has frequently been shortened over time, down to a duration of 10 months in the 1990s. In February 1996, Jacques Chirac, the president of France at the time, declared his intention to abolish mandatory military service in order to professionalize it. The French government began discussing the reform of the military service in November 1996, and the law was voted and passed in October 1997. It aimed at progressively suspending compulsory military service, seeking to end it completely by 2003. During this transition period, the former regulations on military duty continued to apply to men born before 1979, even if they postponed their conscription, while those born after December 31st, 1978 were exempt from compulsory conscription. A decree published in June 2001 officially marked the end of mandatory military service for all in France.

2.2 Naturalization

Unlike the United States, most migrant-receiving countries have not adopted unconditional birthright citizenship, or the *jus soli* principle (right of soil). In France, children born on the French territory to two foreign parents can obtain French citizenship prior to reaching the age of 18 upon request, or automatically at the age of 18.^{3,4} Those who do not wish to acquire French citizenship at 18 years old can renounce the automatic naturalization through the “voluntary loss of nationality”. To do so, individuals must submit the required documents to the Court of Justice between the ages of 17 and a half,

¹These 13 countries are Austria, Belgium, Denmark, France, Germany, Ireland, Italy, Luxembourg, Netherlands, Norway, Spain, Sweden, and the United Kingdom. See <https://www.coe.int/en/web/conventions/full-list?module=signatures-by-treaty&treatynum=043> for an overview of the signature and ratification dates.

²The Franco-Algerian convention allows bi-nationals to choose the country in which they would like to complete their military service, irrespective of their place of residence. However, we show that the hardship of the home country’s military service is not relevant in our analysis.

³The majority of children obtain citizenship at age 18. They can also obtain it as early as 13 years of age if the child was born in France and has resided in the country since the age of 8, upon the request of parents. Above the age of 16, children can apply for French citizenship if they have resided in France for at least 5 years since the age of 11.

⁴The rules were changed between 1993 and 1998. The Pasqua Law of 1993 required children born in France to foreign parents to formally declare their wish to acquire French citizenship to obtain it. Other than that, the acquisition remained automatic. This reform should not affect our setting because it should not affect men and women differently. Indeed, we show that the naturalization rate of males and females did not evolve differentially at the moment of the reform.

and 19.⁵ An estimated 10 to 15% of young adults renounced citizenship in the 1980s (Massot, 1985). If they chose to renounce their French citizenship, they could not be reinstated due to the unfulfillment of military obligations (Spire and Thave, 1999) prior to its abolition. In all cases, they would be subject to the more stringent requirements faced by other foreigners if they subsequently wished to acquire French citizenship. These made the refusal at 18 an almost definite choice.

2.3 Naturalization and Conscription

Historically, naturalization policies have been tightly related to military needs. Indeed, the law of the 26th of June 1889, which established the automatic attribution of French citizenship to children of foreign parents born in France, can be traced back to demographic and military concerns (Massot, 1985). It was notably put in place to ensure that second-generation immigrant men would be liable to serve in the French military (Spire, 2005).

Conscription thus became an obligation associated with obtaining citizenship. The timing of the choice to naturalize coincided with the enrollment for military service, making the latter salient in the naturalization decision. In fact, it is estimated that around 90% of individuals who renounced citizenship among second-generation in 1990s were men (Weil, 2002). This clearly suggests that military service was considered as a cost of naturalization. Thus, the abolition of compulsory conscription in 1997 made naturalization less costly for second-generation men. Given the delay between the first presidential announcement in February 1996 and the adoption of the reform in October 1997, we document a slight anticipation effect in Figure A.15, and account for this in a robustness analysis.

Second-generation men who obtained the French nationality while retaining their origin countries' nationality, presumably fulfilled their military service obligation in the country of usual residence. We address concerns that the naturalization take-up could be a function of the ease of the origin country's military service in a robustness check in Section 6.

3 Data

3.1 French Population Census

We conduct the main analysis on the French Population Census, collected by the French National Institute for Statistics and Economic Studies (INSEE). Until 1999, it was exhaus-

⁵The documents consist of a copy of the birth certificate and identity card, a recent photo, a proof of another country's nationality, a proof of no participation in the French military service, and a refusal declaration.

tive and collected every 6 to 9 years. Since 2004, it has been exhaustive for municipalities under 10,000 inhabitants and it has covered 40% of dwellings in municipalities that exceed this threshold. We thus weight all computations using the survey weights provided by INSEE. Under this new data collection procedure, 20% of individuals are surveyed each year such that each period of 5 consecutive yearly census surveys constitutes a complete survey wave.

We specifically rely on the 2014 survey wave, which gathers forms collected from 2012 to 2016, such that individuals born around January 1st 1979 are surveyed at age 35 on average. French citizenship take-up is thus observed relatively early in adulthood, and labor market outcomes are observed sufficiently late for individuals born after 1978 to have completed their education and entered the labor market. Note that the collection process of a complete survey wave over 5 years generates some age variation within birth cohort, which we can control for in the analysis.

The census notably collects information on individuals' age, gender, educational and labor market outcomes, marital status and nationality. For French citizens, both the current nationality and the birth nationality are observed. This allows us to distinguish individuals who acquired the French citizenship from those who were born French. For non-French citizens, the current nationality is observed.⁶ The census does not contain information on dual nationalities.⁷

Information on family links between censused individuals is only available for 40% of the full sample. We exploit this feature of the data to study the outcomes of the spouse. Hereinafter, we refer to this subsample as the *complementary census sample*, while we refer to the full sample as the *main census sample*.

3.2 Permanent Demographic Sample

We rely on the Permanent Demographic Sample (EDP) to investigate attrition patterns and the timing of citizenship acquisition over the life course. The EDP is a panelized version of the census data for a subset of individuals born on specific dates during the year.

Specifically, it includes individuals born during the first 4 days of April, July, October, and from the 2nd to the 5th of January. Thus, the EDP covers about 4% of the population. Census information of EDP individuals is observed at variable ages and calendar years due

⁶For these individuals, we must assume that their birth nationality is identical to their current nationality. We test this assumption using the Permanent Demographic Sample, described in Section 3.2, which follows a sample of the population across census waves. In this sub-population, 98% of the foreigners born ± 10 years around the cutoff, i.e., from 1969 to 1998, and observed in both the 2014 census and in the 1990 census, have the same nationality in the two census waves.

⁷As long as an individual was born French, no information on other potential nationalities is available. For those who acquired French citizenship, a single birth nationality can be reported. For those who were not French when they were surveyed, a single current nationality is reported even if they had more than one nationality.

Table 1: Descriptive statistics of the main sample

	Male	Female	All
Age	35.05	34.59	34.81
Citizenship acquisition	72.18%	84.22%	78.37%
High school	55.12%	68.96%	62.24%
Higher education	35.34%	46.60%	41.13%
Birth nationality:			
European Union	51.45%	46.80%	49.06%
Other Europe	11.65%	11.93%	11.80%
Africa	31.00%	35.09%	33.11%
Other	5.90%	6.18%	6.04%
Observations	65,347	67,722	133,069

Notes: This table provides the average demographic characteristics of our sample of interest: individuals born in France without French citizenship from 1969 to 1988. European Union is defined as in 1996. A more detailed breakdown of the birth nationality composition of the sample is provided in Table B.1. Source: *French Population Census*, wave 2014, INSEE.

to the rotating yearly collection procedure of the French population census. On average, individuals are observed once every 5 years in the EDP census data.

Census information of EDP individuals is also merge with data from the electoral register. This is of particular interest because it contains the date at which individuals were added to the electoral register, which we can exploit to impute the age at which individuals acquired citizenship.

3.3 Sample definition

Our sample of interest consists of second-generation immigrants, i.e., individuals born in France without French citizenship. They represent 1.3% of the population among individuals born 10 years around the threshold, from 1969 to 1988. Table 1 shows the descriptive statistics of this population using the main census sample.

This sample comprises around 6,500 individuals per birth cohort, with about as many men as women. Individuals are observed at age 35 on average, and 78% of them acquired French citizenship. Females tend to be more educated than males, and they take up French citizenship more often. Almost half of the sample has a birth nationality that pertains to the European Union, and African birth nationalities represent a third of the sample. The shares of the most represented birth nationalities in our sample are presented in Appendix Table B.1.

4 Compulsory conscription and naturalization

4.1 Theoretical framework

4.1.1 General framework

We present a simple theoretical framework to describe the relationship between the costs associated with citizenship and the take-up of citizenship. We consider a simple setting where an individual i would take up citizenship as soon as the corresponding benefits B_i exceed the associated costs C_i . Individuals are characterized by their skill level S_i , distributed over the interval $[s^- \geq 0, s^+]$. Typically, costs imposed on citizenship such as language tests, financial requirements, or even simply administrative burden, are decreasing with skills. Thus, individuals' take-up decision can be modeled as:

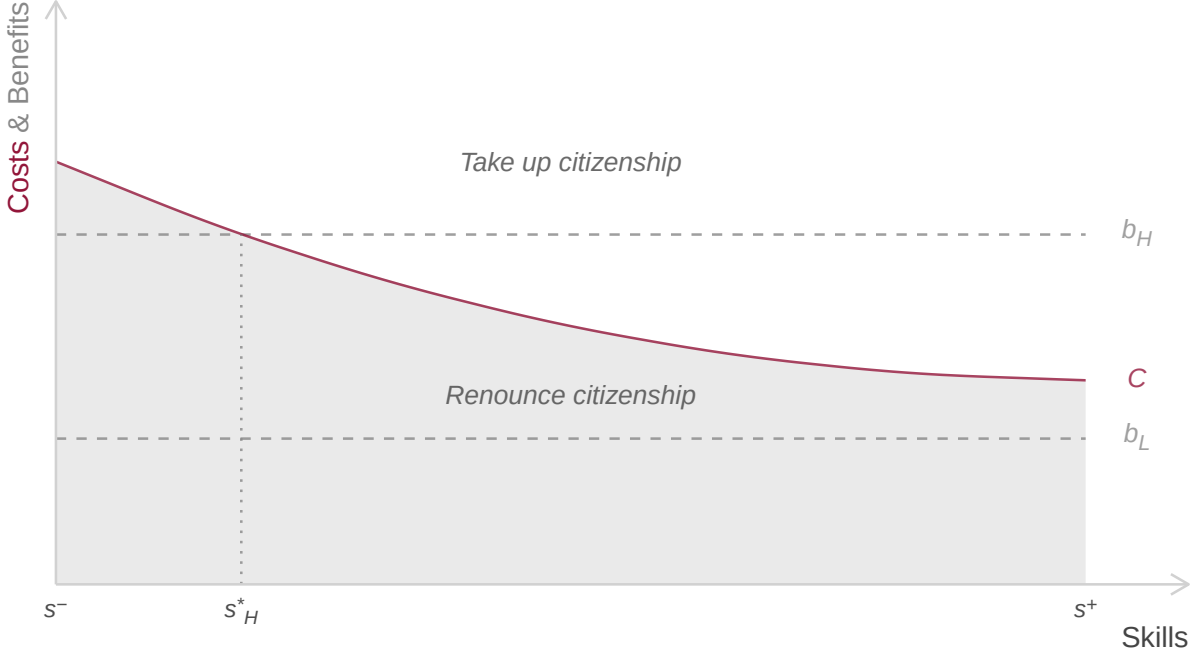
$$\text{Take-up}_i = \mathbb{1}\{B_i \geq C(S_i)\}, \text{ with } \frac{\partial C}{\partial S} < 0.$$

Homogeneous benefits. To begin with, consider that $B_i = b \forall i$. If $b > C(s^-)$, everyone would take up citizenship. If $b < C(s^+)$, no one would take up citizenship. For any intermediate value of b , there is a skill level s^* such that costs and benefits equalize: $b = C(s^*)$. In this situation, only individuals whose skill level S_i is greater than the threshold s^* would take up citizenship. Thus, when benefits are homogeneous, the higher the costs associated with citizenship, the higher the skills of individuals who take up citizenship.

Heterogeneous benefits. In most contexts, all individuals do not benefit from citizenship to the same extent. For instance, benefits may be larger for individuals whose birth nationality is discriminated against on the labor market. To account for that, let the benefits derived from citizenship vary with individuals' type $k \in (L, H)$ such that $b_L < b_H$. The higher the level of benefits b_k derived from citizenship, the lower the skill threshold s_k^* above which individuals would take up citizenship: $s_H^* < s_L^*$.

In this situation, if high-benefits individuals tend to be lower-skilled than low-benefits individuals, shifting the cost function upwards would not necessarily lead to select the top of the skill distribution. Consider for instance the case depicted in Figure 1. The cost function C is such that $C(s^+) > b_L$ screening out every high-skill low-benefits individual. Given that $s_H^* > s^-$, the lower end of the skill distribution of low-skill high-benefits individuals is also screened out. Overall, such a cost function would lead to an overrepresentation of the middle part of the skill distribution among individuals who acquire citizenship. A flatter cost function could even be such that $C(s^-) < b_H$, selecting every individual from the low-skill group but none from the high-skill group.

Figure 1: Citizenship take-up decision under heterogeneous benefits



Notes: This figure shows a schematic depiction of our theoretical framework. Costs and benefits of citizenship acquisition are represented on the y -axis. The skill level of individuals confronted with the choice of either taking-up or renouncing citizenship varies along the x -axis, from s^- to s^+ . The solid curve C represents a typical cost associated with citizenship acquisition. Any individual whose level of benefits is below this curve would renounce citizenship. The horizontal dashed lines b_L and b_H represent benefits that individuals from the low-benefits group and the high-benefits group respectively derive from acquiring citizenship. As b_L lies below C over the whole $[s_H^*, s_L^*]$ interval, no one from the low-benefits group would take up citizenship. Individuals from the high-benefits group whose skill level is below s_H^* would not take up citizenship either. Only individuals from the high-benefits group whose skill level exceeds that threshold would take up citizenship.

4.1.2 Application to the French context

In the 1980s, 10 to 15% of individuals born in France to foreign parents did not take up citizenship despite the automatic acquisition process (Massot, 1985). A potential explanation to this phenomenon is that compulsory conscription disincentivized the offspring of immigrants to take up French citizenship.

The cost of doing military service is likely to decrease with education, which is reminiscent of actual naturalization policies whose costs are usually decreasing with skills. Indeed, less strenuous positions were typically assigned to more educated conscripts (Maurin and Xenogiani, 2007). In addition, lower educated individuals tend to be more liquidity constrained as they usually come from lower socio-economic backgrounds. They may need to work to earn money for them and their family, which is not compatible with military duty.⁸ Military service may also have beneficial aspects, such as exposition to social di-

⁸The fact that higher-educated individuals would have higher forgone expected earnings increases the relative cost of military service at the top of the education distribution, but we do not expect this effect to be strong enough to reverse the sign of the relationship.

versity, or the possibility to obtain a driver’s license for free (Avrillier et al., 2010). We assume that these potential benefits are homogeneous and we implicitly include them in the costs, which must be interpreted as the *net* cost of military service. Given the automatic acquisition process, we consider that other costs are negligible in our framework.

Benefits of acquiring French citizenship notably include electoral rights and access to public sector jobs restricted to the French population.⁹ A strong dimension of heterogeneity in the benefits of citizenship is the birth country. Since 1958, the Treaty of Rome allows European Union citizens to freely reside and work in any country of the EU regardless of their specific nationality. Individuals whose birth nationality does not pertain to the EU would thus derive larger benefits from French citizenship in terms of labor market access and residence rights. On top of that, individuals whose birth nationality is typically discriminated against in the labor market, such as African nationalities (Duguet et al., 2010; Adida et al., 2016; Primon and Simon, 2018; Govind and Santini, forthcoming), would benefit even more from French citizenship as a signal to employers.

In this context, the decision to take up citizenship can be modeled as follows.

$$\text{Take-up}_i = \mathbb{1}\left\{ B \underset{(+)}{\left(\mathbb{1}\{\text{Birth nat}_i \notin \text{EU}\}, \mathbb{1}\{\text{Birth nat}_i \in \text{Africa}\} \right)} + \underset{(-)}{\eta_i} \geq C(\text{Education}_i) \right\},$$

where η_i represents exogenous and normally distributed individual preferences for acquiring French citizenship that generate variation in the take-up decision for a given birth nationality and education level. Note that the cost of military service C only applies to males whose birth cohort was subject to compulsory conscription, and may not be binding to all birth nationality groups. We can draw three main hypotheses from this theoretical framework.

First, if the cost of military service is sufficiently large, the share of males born in France to foreign parents who acquired French citizenship should increase with the abolition of compulsory conscription, while that of females should remain constant.

Second, this increase should be larger for less educated men. The cost of military service is expected to be larger for them, and it is thus more likely to be binding.

Third, this increase should be larger for men who were born citizens of the European Union than for men born with other nationalities. The latter have larger benefits of acquiring French citizenship, especially individuals with an African nationality who are typically discriminated against on the labor market.

4.2 Empirical framework

We expect the abolition of compulsory conscription to reduce the disincentive for males born in France without French nationality to take up French citizenship at the age of 18.

⁹Note that since the law n° 91-715 of July 26, 1991, 80% of public sector employment are accessible to EU citizens.

To test this hypothesis we exploit the fact that women were not subject to compulsory conscription. Specifically, we compare the difference in naturalization rates for men born before and after 1979 to the difference in female naturalization rates in a Difference-in-Differences setting. Equation 1 displays the corresponding specification.

$$\text{Naturalization}_i = \alpha + \beta \text{Male}_i + \delta \text{Post}_i + \gamma(\text{Male}_i \times \text{Post}_i) + \varepsilon_i \quad (1)$$

Naturalization_i is a dummy variable taking the value 1 if the individual acquired French citizenship and 0 otherwise. Male_i is a dummy variable taking the value 1 for males and 0 for females. Post_i is a dummy variable taking the value 1 if the individual was born after 1978 and 0 otherwise, since only males born until 1978 were subject to compulsory conscription.

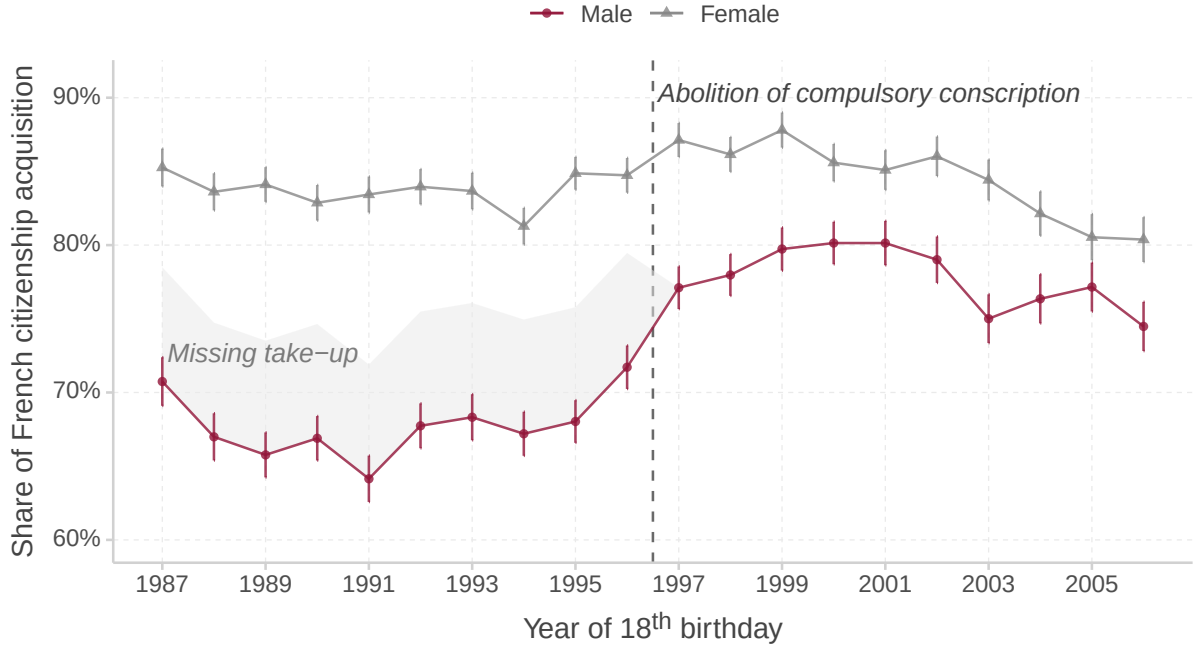
Our coefficient of interest is γ , which quantifies the difference in the naturalization rates for men born before and after 1979, net of the difference in the naturalization rates of women. According to the theoretical framework developed in Section 1, we expect γ to be positive. For γ to capture the causal effect of the abolition of compulsory conscription on naturalization rates, the difference in female naturalization rates between the two periods must be equal to the difference that would have been observed for males absent the abolition of compulsory conscription. For the parallel trend assumption to hold, the naturalization rate of females must not be directly affected by the reform, nor indirectly through their spouses. Also, no event concomitant to the reform must have affected the naturalization rates of men and women. We provide evidence supporting these two assumptions in Section 4.3.

4.3 Results

Figure 2 shows the naturalization rates of individuals born in France without French citizenship by birth cohort and separately for men and women. Given that the trade-off between renouncing French citizenship and doing military service is faced at age 18, birth cohorts are labeled according to the year of 18th birthday on the x -axis. The vertical dashed line represents the moment of the abolition of compulsory conscription. The corresponding difference-in-differences regression results for a window of ± 5 years around the threshold are reported in Appendix Table B.2.

The naturalization rate of females remains stable at around 85% over the period. The fact that it never reached 100% even though females were not subject to compulsory conscription could be explained by factors such as exclusive nationalities, identities and sense of belonging, individual preferences, the intention not to stay in France, or not fulfilling the residence requirement. The naturalization rate of males, however, does increase markedly at the threshold. This is in line with the fact that the automatic acquisition of French citizenship was suddenly no longer tied to the cost of doing military

Figure 2: Naturalization rates of second-generation immigrants



Notes: This figure represents the share of French citizenship acquisition among individuals born in France without French citizenship, separately for males (red) and females (gray), for birth cohorts from 1969 to 1988. Vertical lines show the corresponding 95% confidence intervals. The x -axis is labeled according to the year of 18th birthday, which is when males born before 1979 must decide whether to do military service or to renounce French citizenship. The vertical dashed line represents the abolition of compulsory conscription. For males, after this point, citizenship acquisition is not tied to doing military service anymore. The shaded area represents the estimated fraction of missing citizenship take-up among young males caused by compulsory conscription. The height of the area is obtained from a Difference-in-Differences regression between males and females born 5 years before and after 1979, as specified in Equation 1. The corresponding regression results are shown in Appendix Table B.2. Source: *French Population Census*, wave 2014, INSEE.

service, which increased take-up.

We estimate the overall share of compliers with a Difference-in-Differences following Equation 1. It amounts to 7.7% and is represented graphically by the shaded area. Corresponding regression results are reported in Appendix Table B.2. Thus, the abolition of compulsory conscription halved the initial gap of 15.2% in naturalization rates between males and females. As the take-up rate for males born before 1979 amounts to 68.5%, this result suggests that compulsory conscription was responsible for about a quarter of the missing take-up of French citizenship among young males before its abolition. In absolute terms, more than 4,500 young males born in France renounced French citizenship because of compulsory conscription during the decade leading to its abolition.

4.4 Assumptions validity

This result can only be interpreted as causal if the trend in the naturalization rate of females is a valid counterfactual for that of males. The fact that the two trends are parallel

in the pre-period supports the validity of this assumption. The p-value associated with the 5-year pre-trend is 0.39. Still, this does not rule out that naturalization rates of males and females may have diverged absent the reform. We must ensure that the reform had no indirect impact on females, and that no concomitant event may confound its effect.

4.4.1 Absence of indirect effect on females

Even though females are not directly concerned by the abolition of compulsory conscription, those who married someone affected by the reform could in turn be affected through the possibility of acquiring citizenship after 4 years of marriage with a French person. We test this hypothesis by plotting the naturalization rate of females against the birth cohort of their husbands instead of their own. Because these computations require identifying family links between censused individuals, we rely on the complementary census sample instead of the main census sample.

Results are displayed in Appendix Figure A.1. Among women whose husband was born in France, we distinguish women whose husband was born French from women whose husband was born foreigner. In both cases, no discontinuity or kink in naturalization rates is observed between the two periods. This suggests that females were not indirectly affected by the reform through their spouses.

4.4.2 Absence of concomitant events

Any event differently affecting males' and females' naturalization rates simultaneously to the reform could threaten the validity of our estimates. To the best of our knowledge, no reform other than the abolition of compulsory conscription have affected differently males and females born before and after 1979. However, two events that may have impacted citizenship take-up occurred close to 1997, when the reform passed.

The first one is the inclusion of Austria, Finland, and Sweden in the European Union in 1995. After this change, citizens from these countries could work and reside freely in France, which lowered their incentive to acquire French citizenship. However, this should not affect males and females differently. In addition, these birth nationalities account for only 0.12% of our sample, so their exclusion cannot threaten the robustness of our result given its magnitude.

The second event is the Pasqua law, implemented in 1993 and abolished in 1998. It required individuals to formally declare their wish to acquire French citizenship to obtain it. Other than that, citizenship acquisition remained automatic. In Figure 2 we observe a slight drop in naturalization in 1993 for both men and women, which did not seem to induce any divergence between the two trends. The abolition of the Pasqua law in 1998, one year after the abolition of compulsory conscription, does not seem to have affected either of men's and women's trend in naturalization rate.

4.5 Heterogeneity

Following the theoretical framework outlined in Section 4.1, we then investigate the heterogeneity of this effect according to education and birth nationality group. Specifically, we distinguish individuals who studied up to high school from those who pursued higher education. Within these two groups, we also distinguish individuals whose birth nationality pertains to the European Union from other European nationalities, and from African nationalities.¹⁰ The remaining nationalities are very heterogeneous and would form a too small group for their naturalization rates to be reliably computed by gender and birth cohort.

Figure 3 shows the evolution of naturalization rates by birth cohort for each of the 6 resulting groups, for males and females separately. Corresponding regression results are shown in Appendix Table B.3. Female naturalization rates are relatively stable over the period irrespective of the education level and of the group of birth nationality. Male naturalization rates are generally lower, and they do not follow the same trend depending on the group of birth nationality. The naturalization rate of males born with a non-EU nationality, either European or African, follow a very similar trend to that of females. For individuals born EU citizens, however, there is a sudden increase in naturalization rates at the moment of the reform. This clearly shows that the impact of the abolition of compulsory conscription is entirely driven by EU citizens. This is in line with the incentive scheme described in Section 4.1, as this group is the one for whom the benefits of acquiring French citizenship are expected to be the lowest.

Also, within European Union citizens at birth, we observe that the effect is more than 50% larger for the lower educated (12.7 pp., against 7.7 pp. for the higher educated). This is consistent with the idea that the cost of military service is much larger for the low-educated, and thus more likely to be binding for this subgroup.

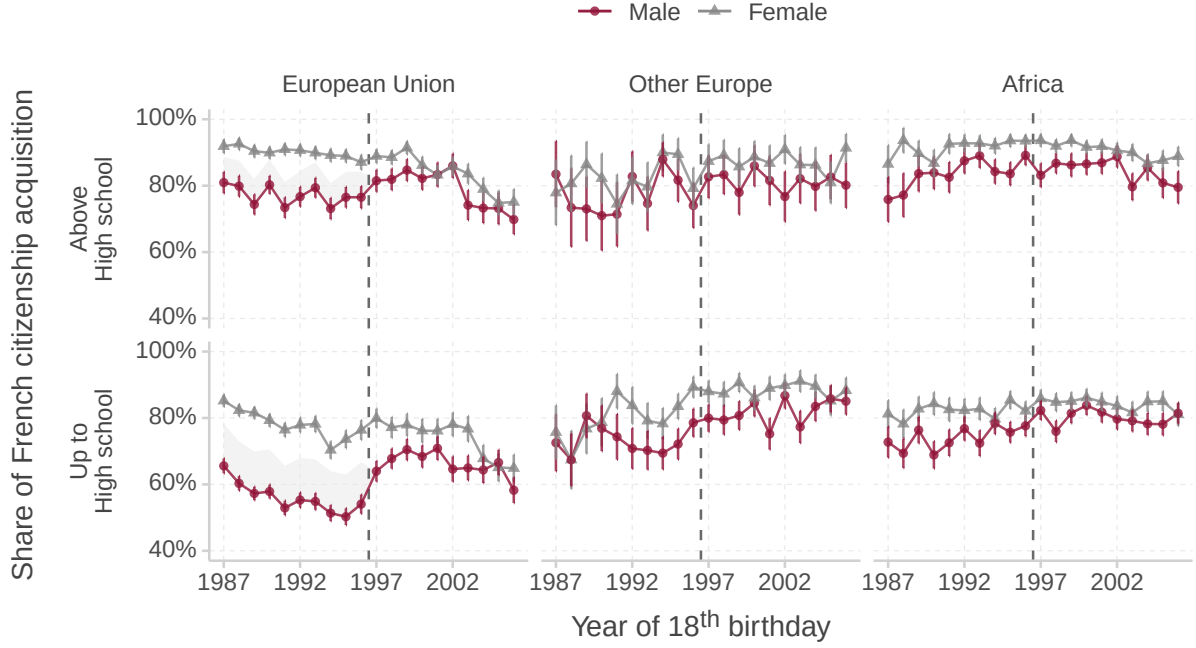
5 Naturalization and labor market outcomes

5.1 Empirical framework

We exploit the sudden increase in naturalization among males born in France without French citizenship to estimate the causal impact of naturalization on labor market outcomes. As shown above, the only group to react to the abolition of compulsory conscription was males born EU citizens, while other groups of birth nationalities remained

¹⁰We consider European Union as in 1996, i.e., Austria, Belgium, Denmark, Finland, Germany, Greece, Ireland, Italy, Luxembourg, the Netherlands, Portugal, Spain, Sweden, and the United Kingdom. Other European countries are grouped into the non-EU category. Because Austria, Finland, and Sweden entered the EU only two years prior to the reform, we provide robustness checks without these three countries in Appendix Figure A.2 and Appendix Table B.4. Results remain virtually unaffected by the exclusion of these countries.

Figure 3: Naturalization rates by education and birth nationality groups



Notes: This figure represents the share of French citizenship acquisition among individuals born in France without French citizenship for 6 subgroups defined according to birth nationality (European Union, Other Europe, and Africa) and education (up to high school and above high school). European Union is defined as it was in 1996. In each panel, shares of citizenship acquisition are represented separately for males (red) and females (gray), for birth cohorts from 1969 to 1988. Vertical lines show the corresponding 95% confidence intervals. The x -axis is labeled according to the year of 18th birthday, which is when males born before 1979 must decide whether to do military service or to renounce French citizenship. Shaded areas represent the fraction of missing citizenship take-up among young males caused by compulsory conscription. The height of the area is obtained from Difference-in-Differences regressions between males and females born 5 years before and after 1979, estimated on the corresponding subgroups as specified in Equation 1. Regression results are reported in Appendix Table B.3. Significant effects are found only for individuals born citizens of the European Union. Source: *French Population Census*, wave 2014, INSEE.

unaffected by the reform. Thus, several potential control groups can be considered.

Females born EU citizens belong to the same group of birth nationalities as the treated group, but they may not share the same trend in labor market outcomes due to the gender difference. Males born without the EU citizenship have the same gender, but birth nationality differences may induce differences in trends. Still, these different potential control groups all have common characteristics with the treated group that are likely to induce co-movement in the outcomes of interest. It makes our empirical setting suited for the use of a Synthetic Difference-in-Differences strategy (Arkhangelsky et al., 2021).¹¹ This approach extends the Synthetic Control approach introduced by Abadie and Gardeazabal (2003) in two ways.

First, the inclusion of a group fixed effect allows us to weight control groups based on

¹¹We cannot conveniently implement a Triple-Difference nor a Regression Discontinuity design in our setting due to the lack of statistical power provided by our sample.

their demeaned trend instead of their absolute trend. Thus, a control group that is more distant from the treated group in absolute terms but whose time variations are similar can be attributed a higher weight.

Second, the Synthetic Difference-in-Differences approach introduces time weights for pre-treatment years in addition to group weights.¹² Time weights are computed to give more importance to time units for which the outcome levels of control groups are closer to their average in the post-period. This typically gives more importance to the fit of the most recent time units in the pre-period to prevent potential discrepancies right before the event from contaminating the estimated treatment effect.

The estimated model is as follows.

$$(\hat{\tau}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\tau, \mu, \alpha, \beta} \left\{ \sum_{g=1}^G \sum_{t=1}^T (Y_{gt} - \mu - \alpha_g - \beta_t - W_{gt}\tau)^2 \hat{\omega}_g \hat{\lambda}_t \right\}, \quad (2)$$

where the G groups are indexed by g , and the T birth cohorts are indexed by t . Y_{gt} denotes the average of a given labor market outcome among individuals from group g and birth cohort t . μ is a constant term, α_g a group fixed effect, and β_t is a time fixed effect. W_{gt} is a binary variable taking the value 1 for the treated group in post-reform time units, and the value 0 otherwise. $\hat{\omega}_g$ and $\hat{\lambda}_t$ are the vectors of group and time weights, computed following [Arkhangelsky et al. \(2021\)](#). Each of these weight vectors sums to 1 and can only take positive values. Note that with a single control group and two periods, weights do not matter and this approach boils down to the standard Difference-in-Differences specification. Similarly, the standard Synthetic Control specification can be obtained by omitting the group fixed effect α_g and the vector of time weights $\hat{\lambda}_t$.

Our set of control groups includes all female birth nationality groups, i.e., females born EU citizens, females born with a non-EU European nationality, and females born with an African nationality. It also includes male birth nationality groups for whom the cost of military service was not binding, i.e., males born with a non-EU European nationality and males born with an African nationality. To enlarge our set of control groups we also include females and males born abroad with a nationality outside the EU and who arrived in France early during childhood.¹³ In particular, for comparability with individuals born in France from foreign parents, we consider children who arrived in France by the age of 6 to ensure that they were schooled in France from the beginning of primary school. We also exclude EU citizens from the group of males born abroad because even though they are not subject to the automatic acquisition of citizenship, the cost of doing military service may enter their decision to apply for citizenship.

¹²Note that group weights and time weights are computed independently from one another.

¹³In practice, low sample size prevents us from breaking down these two groups even further by groups of birth nationalities. We do not consider individuals born French as potential control groups because of the direct effect of compulsory conscription on education identified by [Maurin and Xenogiani \(2007\)](#). This issue is discussed in more detail in Section 5.3.1.

The correct identification of the parameter of interest τ relies on the parallel trend assumption. Specifically, the trend followed by the synthetic control group, i.e., by the weighted average of the different control groups considered, must be the same as the trend that the treated group would have followed absent the shock. This requires (i) that the abolition of compulsory conscription must not have impacted the outcome of the treated group other than through the jump in naturalization it induced, (ii) that no concomitant event has impacted differently the treated group and the control groups, and (iii) that none of the individual control groups were impacted by the reform, neither directly nor indirectly. We provide support for the validity of these assumptions in Section 5.3.

Under these assumptions, the parameter τ identifies an intention-to-treat (ITT) impact. Indeed, it captures the average effect on all EU males regardless of compliance status. The average treatment effect on the complier population, i.e., the local average treatment effect (LATE), can be obtained by dividing the ITT by the share of compliers. The share of compliers corresponds to the baseline Difference-in-Differences coefficient γ from Equation 1 estimated on EU citizens at birth. Regression results for individuals born within 5 years before or after 1978 are presented in Appendix Table B.5, with and without controls for age and education. The share of compliers amounts to 11.9%. Yearly Difference-in-Differences coefficients are shown in Figure A.3. The Synthetic Difference-in-Differences estimate of the share of compliers amounts to 10.7%. Corresponding results are shown Appendix Figure A.4 and Appendix Table B.6.

5.2 Results

We implement the Synthetic Difference-in-Differences methodology to estimate the effect of the abolition of compulsory conscription on employment, unemployment, and inactivity for EU males through the jump in naturalization only them experienced.

Figure 4 provides a graphical representation of the results obtained for the probability of employment. The top panel displays the average share of employed individuals by birth cohort separately for each group. The solid red line corresponds to the evolution of the outcome of EU males, those who experienced a jump in their naturalization rates at the moment of the reform. The individual trends underlying that of the synthetic control group are reported in light gray separately for each control group.

Over the whole period, EU males have the most favorable employment outcomes. EU females do also have good employment outcomes relative to other groups. This highlights the importance of relying on the Synthetic Difference-in-Differences approach instead of the standard Synthetic Control in our setting. Indeed, while the latter would give virtually all the weight to EU females despite slightly converging trends, the former abstracts from the proximity in levels and it weights groups based on their demeaned trend. The resulting synthetic trend, computed following [Arkhangelsky et al. \(2021\)](#)'s methodology,

and represented with a dark gray line, closely follows that of the treated group, with more stability than any other individual trend. This makes Synthetic Difference-in-Differences also more suited than standard Difference-in-Differences in our setting.

The dashed red line represents the counterfactual evolution of the treated group absent the reform. It is obtained by shifting the synthetic trend by the average difference between the trend of the treatment group and that of the synthetic control group in the pre-period. The vertical dashed line represents the implementation of the reform. Before the reform, the synthetic control group and the treated group share the same slightly decreasing but steady trend in employment. While the synthetic control group remains on the same path after the reform, a jump in employment is observed for the treated group.

The bottom panel displays the difference between the treated group and the synthetic control group with a solid red line. It is centered at the pre-treatment average difference, weighted by the time weights $\hat{\lambda}_t$. Vertical lines show the corresponding bootstrapped 95% confidence intervals. The gray lines are placebo effects, each using one of the control groups as the treatment group instead of EU males. While all placebo differences in trends remain relatively gathered around 0, the actual difference in trends exhibits a clear upward shift at the moment of the reform.

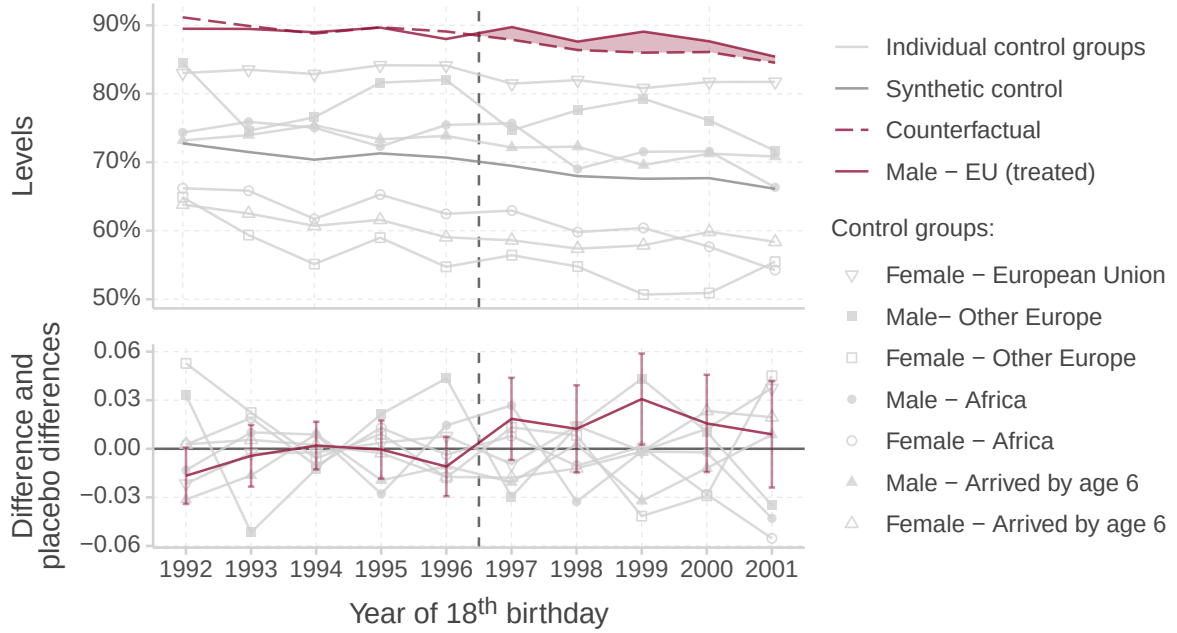
Table 2 column (1) shows the baseline Synthetic Difference-in-Differences results on employment, controlling for age and education. Columns (2) and (3) report the results for unemployment and inactivity. Corresponding graphical representations are presented in Appendix Figures A.5 and A.6. The first row shows the intent to treat effect, estimated as specified in Equation 2 controlling for age and education. The second row documents the corresponding local average treatment effect. It is computed as the ITT divided by the share of compliers, which is estimated as specified in Equation 1 controlling for age and education. Bootstrapped standard errors are reported in parentheses.

Columns (1) indicates a significantly positive effect of naturalization on the probability of employment. The magnitude of the ITT amounts to 1.7 percentage points, which is substantive given that it is driven by the 12% of the treated group who actually acquired citizenship in response to the abolition of compulsory conscription. Indeed, the LATE indicates that the effect on compliers reaches 14.5 percentage points. Given that the probability of employment in the treatment group is around 90%, as can be seen from Figure 4, this suggests that among EU males, compliers have lower employment rates than average. This is in line with the fact that the lower educated are those who reacted the most to the reform, as observed in Figure 3.

Columns (2) and (3) suggest that this increase in the probability of employment is related to a greater extent to a reduction in inactivity than in unemployment. Indeed, while the reduction in inactivity reaches 8 percentage points for compliers, a non-significant 5.4 percentage point decrease is observed in unemployment.¹⁴

¹⁴In a standard Difference-in-Differences setting the coefficients associated with the variables Em-

Figure 4: Synthetic Difference-in-Differences - Employment



Notes: The top panel of this figure shows the share of employed individuals by birth cohort within each of the groups used in our Synthetic Difference-in-Differences approach. That of the treated group, male European Union citizens, is represented with a solid red line. That of each individual control group is represented with a gray line, whose markers indicate the corresponding group. The solid gray line shows the trend of the synthetic control group. The dashed red line represents the counterfactual trend of the treated group. It corresponds to the trend of the synthetic control group shifted by the average difference between the trend of the treatment group and that of the synthetic control group in the pre-period. The bottom panel displays the difference between the treated group and the synthetic control group with a solid red line, centered at the pre-treatment average difference weighted by the estimated time weights. Vertical lines show the corresponding 95% bootstrapped confidence intervals. The gray lines are placebo effects, each using one of the control groups as the treatment group instead of EU males. Corresponding regression results are reported in Table 2. Source: *French Population Census*, wave 2014, INSEE.

For each outcome, group weights are quite evenly distributed across individual control groups. No specific group stands out by having systematically high or low weights compared to other groups. Regarding time weights, earlier birth cohorts tend to be given less weight. This is what is typically obtained by design, as emphasized in Section 5.1.

5.3 Assumptions validity

For the estimated coefficients to capture the causal effect of naturalization, the parallel trend assumption must hold between the treated group and the synthetic control group. In our setting, its validity relies on three core hypotheses. First, military service must not have a direct impact on male labor market outcomes other than through naturalization.

employed, Unemployed, and Inactive would mechanically sum to 0, which is not generally the case with the Synthetic Difference-in-Differences approach. The ITT coefficients reported in Table 2 sum to 0.001, which is reasonably close to 0 given the magnitude of the estimates.

Table 2: Effect of naturalization on labor market outcomes

	Employed	Unemployed	Inactive
ITT ($\hat{\tau}$)	0.017** (0.009)	-0.006 (0.008)	-0.010* (0.006)
LATE ($\hat{\tau}/\hat{\gamma}$)	0.145** (0.079)	-0.054 (0.069)	-0.080* (0.052)
Group weights ($\hat{\omega}_g$)			
Female - European Union	0.092	0.109	0.152
Male - Other Europe	0.159	0.151	0.169
Female - Other Europe	0.167	0.162	0.172
Male - Africa	0.104	0.113	0.052
Female - Africa	0.146	0.155	0.150
Male - Arrived by age 6	0.181	0.144	0.172
Female - Arrived by age 6	0.152	0.166	0.133
Time weights ($\hat{\lambda}_t$)			
1992	0.000	0.000	0.120
1993	0.000	0.000	0.000
1994	0.556	0.382	0.788
1995	0.358	0.618	0.000
1996	0.086	0.000	0.091
Age	✓	✓	✓
Education	✓	✓	✓
Observations	112,468	112,468	112,468
Mean dep. var.	0.714	0.167	0.118
First-stage F-stat.		604	
Share of compliers ($\hat{\gamma}$)		11.91%	

Notes: This table reports the results of a Synthetic Difference-in-Differences which estimates the effect of the increase in the naturalization rates among EU males, induced by the abolition of compulsory conscription, on their employment status. Three binary employment statuses are considered as outcome variables: Employment, Unemployment, and Inactivity. The first row displays the Intention-to-treat effect, estimated with the Synthetic Difference-in-Differences approach described in Section 5.1. The second row shows the Local Average Effect, computed as the ITT divided by the share of compliers. Bootstrapped standard errors are reported in parentheses, and statistical significance is reported according to the following symbology. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. The group and time weights computed following [Arkhangelsky et al. \(2021\)](#) are reported for each specification. Education and age are controlled for in each specification. The F-statistic of the first stage and the share of compliers estimated in Appendix Table B.5 are reported at the bottom of the table. Source: *French Population Census*, wave 2014, INSEE.

Second, female labor market outcomes must not be affected by the reform through the outcomes of their husbands. Third, military service must not be binding in the naturalization decision of immigrants arrived by age 6, which are included in our set of control groups. Evidence provided in this section suggest that none of these threats to identification is at

play in our setting, supporting the validity of the parallel trend assumption.

5.3.1 Absence of direct effect of military service on labor market outcomes

To include males whose birth nationality does not pertain to the EU in our set of control groups, we must ensure that there was no direct effect of military service on labor market outcomes for second-generation immigrants. This is crucial to attribute the effect of the reform entirely to the shock in naturalization rates it induced rather than to the abolition of military service in itself. To the best of our knowledge, no consensus has emerged on the direct effect of military service on labor market outcomes in the literature so far. Peacetime conscription is shown to have a negative effect on earnings in the Netherlands and in Denmark (Hubers and Webbink, 2015; Bingley et al., 2020), and either no effect or positive effects in Germany, Portugal, and France (Bauer et al., 2012; Card and Cardoso, 2012; Mouganie, 2020). Part of the heterogeneity in the evidence put forward is probably due to the large differences in the modalities of military services across countries.

To test whether military service had a direct effect on labor market outcomes, we compare the difference in employment trends by gender separately for second-generation immigrants born EU citizens and those born with other birth nationalities. Indeed, while EU males experienced both the abolition of military service and a jump in naturalization rates, non-EU males only experienced the abolition of military service. Appendix Figure A.7 shows the difference in employment rates between males and females per birth cohort, separately for those born EU citizens and for those born with other European nationalities or African nationalities. While no jump is observed between non-EU males and females, a clear deviation from the pre-trend is observed for EU citizens, with a magnitude comparable to that obtained with the Synthetic Difference-in-Differences.

This suggests that military service did not have a direct effect on labor market outcomes for second-generation immigrants who did not experience a jump in naturalization. As there is no reason for EU citizenship to affect how military service in itself impacts labor market outcomes, this absence of effect is likely to hold for EU citizens as well. This is also in line with (Mouganie, 2020) who documents no or positive direct effects of military service on labor market outcomes in France.¹⁵

5.3.2 Absence of effect on female labor market outcomes

Females were not subject to compulsory conscription before the reform, so they were not directly affected by its abolition. Still, women whose spouse pertains to the treatment

¹⁵Mouganie (2020) shows that men from low socio-economic backgrounds do not adjust on the educational margin, and conscription has no labor market effects for that group. In the case of individuals from high socio-economic backgrounds, the two plausible scenarios resulting from their conceptual framework are that conscription has either no or positive direct effects on employment and earnings.

group could be subject to spillover effects. This could arise if there is assortative mating on citizenship status and if females tend to adapt their employment status to their husbands’.

To test this hypothesis, we document the evolution of females’ outcomes according to the birth cohort of their husbands. Specifically, we compare the trend of females whose husband was born EU citizen to that of females whose husband was born with another nationality. We use the latter as a control group for the former because only EU citizens did react to the reform. Thus, for women whose husband is not an EU citizen, no change in labor market outcomes is expected with respect to whether their husband was born before or after the reform. For spouses of EU males, a sudden change in their labor market outcomes concomitant to the reform would indicate the presence of spillover effects.

Because these computations require identifying family links between censused individuals, we perform this exercise on the complementary census sample. Results are presented in Appendix Figure A.8. Around the threshold, no differential trend is observed between females whose husband was born with an EU nationality and females whose husbands were not. This is suggestive evidence that women were not affected by the reform, even those whose husbands were.

5.3.3 Absence of effect on naturalization for immigrants arrived by age 6

Our results show that except males born EU citizens, no group of second-generation immigrants experienced a jump in naturalization rates at the moment of the reform. We include these groups in our set of control groups, to which we add males and females born abroad and who arrived in France by age 6. For them to be relevant control groups, their naturalization decision must not be affected by the abolition of compulsory conscription either.

Appendix Figure A.9 shows the naturalization rates by birth cohort for male and female immigrants who arrived by age 6, separately for those born EU and non-EU citizens. Similarly to what we observe for second-generation immigrants., the trend remains flat around the threshold for every group except EU males. This suggests that first-generation immigrants do account for the cost of doing military service when applying for French citizenship as well, and that this cost was not binding for non-EU immigrants. This supports the validity of including males and females born abroad and who arrived in France by age 6 in our set of control groups, as long as EU citizens are excluded from the male control group.

However, while second-generation immigrants must decide whether or not to take-up citizenship at the age of 18, first-generation immigrants who want to become French choose at which age they apply for citizenship. Given that the French military service was less intensive past certain age thresholds, we must ensure that the abolition of compulsory conscription did not entail a drop in the age of naturalization for first-generation immigrants. The year of naturalization is not available in census data, so we exploit the

electoral information of the Permanent Demographic Sample and use the age at registration in the electoral register as a proxy for the age of naturalization.

As shown in Appendix Figure A.10, the trends in registration rates in the electoral lists reproduce closely the trends in naturalization rates depicted in Figure 2. We show the evolution of the age at registration in the electoral register by birth nationality groups and generation of immigration in Figure A.11. In this Figure we consider first generation immigrants arrived up to age 18 because the date of registration is not available for enough individuals among those arrived by age 6. The average age at registration of first-generation immigrants arrived by age 18 follows a stable path over the period, with no jump or deviation from the trend around the reform. This supports that males and females born abroad and who arrived in France by age 6 are suitable control groups in our setting.

5.4 Mechanisms

To get a better understanding of how naturalization fostered labor market outcomes, we analyze the effect of the shock on employment within certain types of occupations. Specifically, we focus on public sector jobs and on self-employment because a reaction in these segments of the labor market could be indicative of specific underlying mechanisms.

5.4.1 Public sector jobs

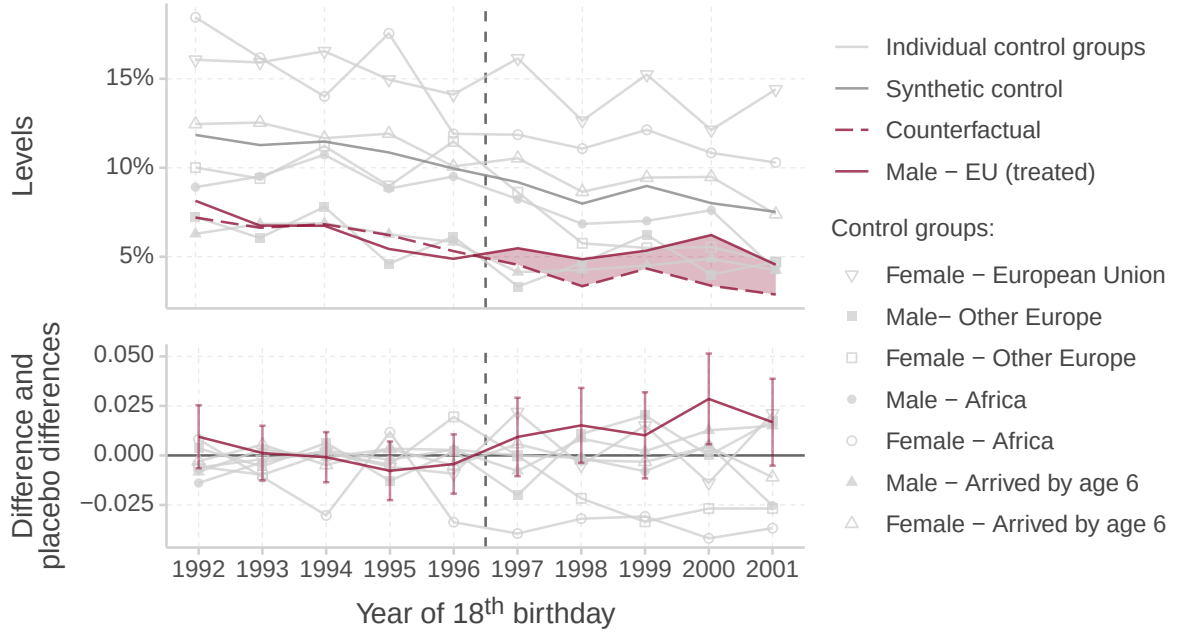
Given that part of the public sector is conditional on having French nationality, naturalization mechanically expands the labor market access of foreigners, providing more outside options. We investigate whether this played a role in the positive employment effect of naturalization. To do so, we estimate a Synthetic Difference-in-Differences as described in Section 5.1 on the share of individuals working in the public sector. Results are represented graphically in Figure 5. The corresponding regression results are documented in Appendix Table B.7.

Results suggest that the 11.9 percentage point increase in the naturalization rates of EU citizens induced a 1.6 percentage point increase in their probability of being employed in the public sector. This corresponds to an 13.3 percentage point increase for the group of compliers. The direction of this effect is coherent with the fact that some public sector jobs require French citizenship. However, the magnitude of the effect is too large to be entirely attributed to these specific positions. This hints at the fact that citizenship is a particularly important criterion in the hiring decision of public sector recruiters.

5.4.2 Self-employment

If the acquisition of citizenship mitigates hiring discrimination, it may provide individuals with more and better labor market opportunities. These new alternative options should

Figure 5: Synthetic Difference-in-Differences - Public job

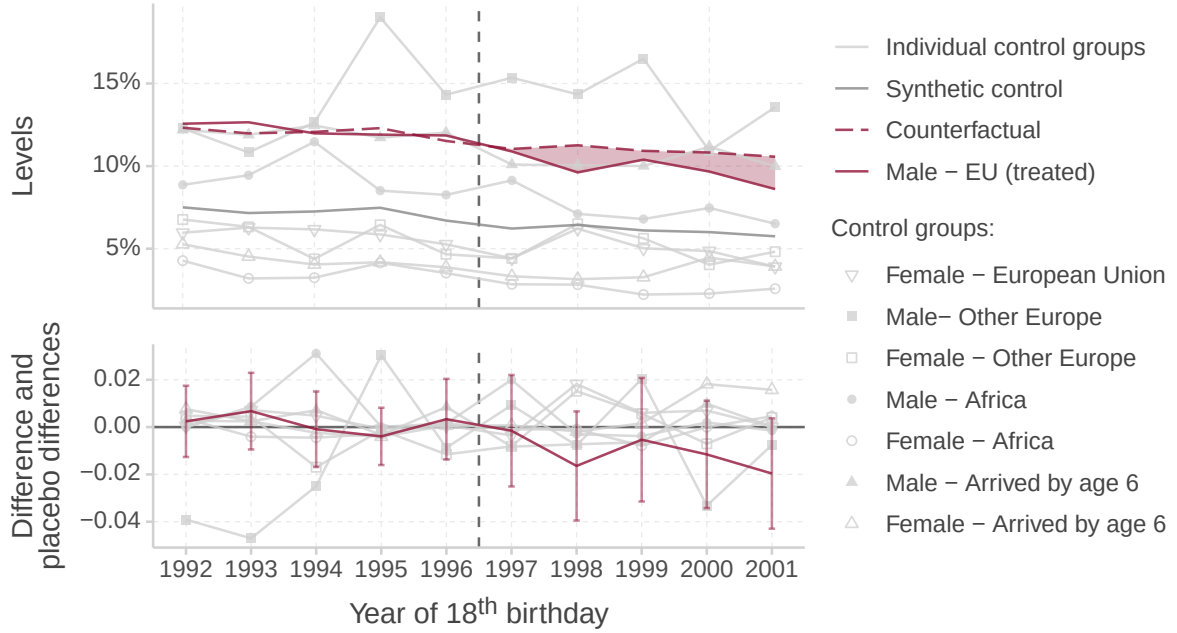


Notes: The top panel of this figure shows the share of individuals working in the public sector by birth cohort within each of the groups used in our Synthetic Difference-in-Differences approach. That of the treated group, male European Union citizens, is represented with a solid red line. That of each individual control group is represented with a gray line, whose markers indicate the corresponding group. The solid gray line shows the trend of the synthetic control group. The dashed red line represents the counterfactual trend of the treated group. It corresponds to the trend of the synthetic control group shifted by the average difference between the trend of the treatment group and that of the synthetic control group in the pre-period. The bottom panel displays the difference between the treated group and the synthetic control group with a solid red line, centered at the pre-treatment average difference weighted by the estimated time weights. Vertical lines show the corresponding 95% bootstrapped confidence intervals. The gray lines are placebo effects, each using one of the control groups as the treatment group instead of EU males. Corresponding regression results are reported in Table B.7. Source: *French Population Census*, wave 2014, INSEE.

reduce the prevalence of more precarious and unstable positions such as self-employment. To test this hypothesis, we estimate a Synthetic Difference-in-Differences on the share of self-employed individuals. Results are represented graphically in Figure 6. The corresponding regression results are reported in Appendix Table B.8.

We observe a decreasing trend in self-employment for the treated group, accentuated after the reform. Even though barely significant, this pattern is in line with the idea that citizenship acquisition could indeed constitute a stepping stone to depart from self-employment towards other types of occupations that may offer more stability on the labor market. This decrease in self-employment could also hint at the fact that citizenship acquisition may lead to a transition from the informal to the formal sector of the economy. This hypothesis is all the more credible given that citizenship acquisition is associated with a significant decrease in inactivity, which would typically be the employment status reported by informal workers (Govind, 2021).

Figure 6: Synthetic Difference-in-Differences - Self-employed



Notes: The top panel of this figure shows the share of self-employed individuals by birth cohort within each of the groups used in our Synthetic Difference-in-Differences approach. That of the treated group, male European Union citizens, is represented with a solid red line. That of each individual control group is represented with a gray line, whose markers indicate the corresponding group. The solid gray line shows the trend of the synthetic control group. The dashed red line represents the counterfactual trend of the treated group. It corresponds to the trend of the synthetic control group shifted by the average difference between the trend of the treatment group and that of the synthetic control group in the pre-period. The bottom panel displays the difference between the treated group and the synthetic control group with a solid red line, centered at the pre-treatment average difference weighted by the estimated time weights. Vertical lines show the corresponding 95% bootstrapped confidence intervals. The gray lines are placebo effects, each using one of the control groups as the treatment group instead of EU males. Corresponding regression results are reported in Table B.8. Source: *French Population Census*, wave 2014, INSEE.

6 Robustness

6.1 Military service in the country of origin

We show in Section 4 that only individuals born citizens of the European Union react to the abolition of compulsory conscription. This is in line with the fact that their benefits of acquiring French citizenship are lower than those of other nationality groups, for which the cost of military service was not binding under compulsory conscription.

Yet, this heterogeneity could also arise from variations in the cost of military service across birth nationality groups. Indeed, for an individual whose country of birth nationality has a military service that is longer or harder than the French one, the French military service would not constitute a comparatively large cost.

We thus distinguish individuals according to whether military service in their country of birth nationality in 1996 was longer or shorter than the French military service, within

the three groups of birth nationality. Appendix Figure A.12 shows the heterogeneity in the trend in male and female naturalization rates along these two dimensions. The effect of the abolition of compulsory conscription does not appear to vary within nationality groups depending on the relative length of the military service. Only European Union citizens at birth do react to the reform, regardless of the relative length of the French military service. This supports the idea that it is indeed a benefits imbalance that drives the heterogeneity across birth nationality groups.

6.2 Anticipation of the reform

Even though the reform did not exempt men born before the 1st of January 1979 from doing military service, it became easier to avoid doing military service after the announcement of the reform.¹⁶ Many of the French men who were born in the last cohorts subject to compulsory conscription, and who postponed their military service to pursue higher education, have actually never fulfilled their military duty.

In our case, anticipation could lead individuals born before 1979, who would not have taken up French citizenship absent the announcement, to actually take up French citizenship because they anticipate that they will be able to avoid military service. This type of behavior would bias our results because these individuals would pertain to cohorts considered unaffected while actually being treated.

To investigate whether this is an issue in our setting, we show the naturalization rates of males and females born EU citizens by month of birth instead of year of birth in Appendix Figure A.15. Due to the amount of noise in the monthly trend, it is difficult to confidently identify from which specific birth month individuals react. Still, there is a clear anticipation phenomenon of two to four months. Anticipation thus starts relatively late given that presidential announcements were made seven and ten months before the effective abolition of compulsory conscription.

It is particularly important to alleviate anticipation effects when using a Synthetic Difference-in-Differences approach because the weighting of time units tends to give more importance to more recent periods, which are the ones potentially subject to anticipation. We thus test the robustness of our result to the exclusion of the 1978 birth cohort, with and without time weights. The resulting ITT coefficients are presented in Appendix Table B.9.

ITT effects tend to be slightly smaller when using time weights, which is usually the case because they tend to fit more closely on recent values of the pre-trend, and thus avoid the propagation of slight wedges at the end of the pre-period to the estimated difference in the post-period. Still, results are robust to the exclusion of the 1978 cohort in terms

¹⁶[Fize and Louis-Sidois \(2020\)](#) document this issue in more detail using data from the archives of the French Ministry of Defense.

of magnitude and statistical significance, both with and without time weights. The same pattern is observed for LATE coefficients, presented in Appendix Table B.11.

6.3 Differential attrition

All along the analysis we implicitly assume that second-generation immigrants would stay in France in adulthood in the same proportions among cohorts turning 18 before and after the abolition of compulsory conscription. If compulsory conscription influences individuals' migration decisions, differential attrition around the threshold could induce a selection bias.

To investigate this potential issue we make use of the Permanent Demographic Sample, which follows a subset of the main population census across census waves. Appendix Figure A.14 shows the probability for individuals born in France without French citizenship to be observed in the 2014 census wave, i.e., in our main sample, conditional on being observed in the 1990 (top panel) or 1999 (bottom panel) census wave by gender and birth cohort. We separate individuals who were born EU citizens from other birth nationalities.

Results reveal no differential attrition around the threshold. Males' and females' probabilities to be observed in the 2014 census wave conditional on being observed in either the 1990 or the 1999 census wave follow a very stable trend over the period. This holds both for individuals whose birth nationality pertains to the EU and for other second-generation immigrants.

6.4 Absence of direct effect of military service on education

Results on the heterogeneous effects of the abolition of compulsory on citizenship take-up across education levels implicitly assume that conscription does not directly impact education choices. However, [Maurin and Xenogiani \(2007\)](#) show that compulsory conscription in France incentivized young males to pursue higher education in order to postpone military service. Hence, the abolition of compulsory conscription had a direct negative effect on males' education compared to females' because education was no longer a way to avoid military service. However, the incentive scheme faced by individuals born in France without French citizenship is not equivalent to that faced by individuals born French. Indeed, most French citizens have no other option than to do military service because they do not have a second nationality to rely on.

To test whether the effect identified by [Maurin and Xenogiani \(2007, Figure 2\)](#) applies to our setting, we reproduce their result as closely as possible with our data.¹⁷ Appendix

¹⁷[Maurin and Xenogiani \(2007\)](#) use the French Labor Force Surveys 1991-2002 to show the changes in the proportion of men and women aged 17 to 23 still in school across birth cohorts. We cannot use the exact same variable definition because of differences in data collection periodicity. Alternatively, we consider two variables: the proportion of individuals who completed high school, and the proportion of individuals who pursued higher education.

Figure A.13 shows the difference between the proportion of graduates in a given birth cohort and the proportion of graduates in the 1974 birth cohort, within gender. We report this separately for individuals born French and for the three groups of birth nationality of second-generation immigrants.

Similarly to [Maurin and Xenogiani \(2007\)](#), we observe that for French-born individuals, the divergence in the growth of the share of graduates between males and females is more marked from 1978 onward than for the previous birth cohort.¹⁸ However, we do not observe this pattern for either of the three birth nationality groups of second-generation immigrants. This suggests that while compulsory conscription did incentivize French-born males to pursue higher education, second-generation immigrants did not adopt this draft-avoidance behavior. This could be explained by the fact that second-generation immigrant males could adjust on the citizenship margin at the age of 18 via their birth nationality.

6.5 General equilibrium effects

The fact that [Maurin and Xenogiani \(2007\)](#) identify a negative effect of the abolition of compulsory conscription on education for the overall male population raises an issue of general equilibrium effect. Indeed, if individuals born after the reform in the majority group become less educated due to the abolition of compulsory conscription, this could constitute a shock on the labor market for other males in these birth cohorts. As the rates of high school graduation and higher education remained quite stable for second-generation male immigrants, their relative education level increased.

To test this hypothesis, we control for the share of natives in individuals' employment zone.¹⁹ Results are presented in Appendix Table B.10. Controlling for the share of natives in the employment zone leaves our estimates virtually unchanged. This absence of a general equilibrium effect could be due to non-substitutability between the two groups resulting from sectoral labor segregation, to labor market frictions in the matching process, or to any other departure from neo-classical settings.

6.6 Control group combinations

We provide a comprehensive assessment of the role that the set of control groups used to generate the synthetic control group has on our results in Appendix Figures A.16 to

¹⁸In our case, however, results indicate that the divergence did not begin in 1978 but started slightly and progressively from the beginning of the study period. This difference with what is put forward in [Maurin and Xenogiani \(2007\)](#) may be due to the difference in the variable definition or to sampling variation between the two data sources. Notably, much more stable trends are obtained from the population census than from the labor force survey, probably because of its much larger sample size.

¹⁹The employment zone is a geographical unit whose borders are based on the share of individuals who both work and live in the zone. It is computed with the algorithm `LabourMarketAreas`, available in `OpenAccess`.

A.20.

Each row of the tile plot on the left panel corresponds to a given set of control group. A tile whose color lies on the gradient in the legend indicates that the control group of the corresponding column was included in the computation of the synthetic control group, while a gray tile indicates that it has been excluded. There are 127 rows, one for each of the possible combinations of control groups formed by sets of 1 control group to 7 control groups. The specification which includes the whole set of 7 control groups is our baseline specification.

Specifications are sorted by the value of the corresponding ITT coefficient reported on the scatter plot on the right panel. The color of each dot indicates the p-value of the pre-trend. To obtain this value, we compute the difference between the average outcome in the treated group and that in the synthetic control group, and regress it on the birth cohort for the years before the reform. The p-value is then computed from a standard two-sided t-test on the slope coefficient. The vertical solid line represents our baseline coefficient, and a vertical dashed line is placed at 0. The kernel density of this distribution of 127 coefficients is shown on the top panel.

There are two main takeaways from these figures. First, pre-trends are quite well captured as long as there are enough control groups in the set. Second, when the set of control groups is large enough to allow the synthetic control group to capture the pre-trend, resulting estimates remain relatively close to our baseline.

7 Conclusion

This paper aims to shed light on the effect of policies that change naturalization costs on second-generation immigrants' naturalization decisions, and on the causal effect of naturalization on labor market outcomes. We rely on the fact that citizenship is automatically granted to individuals born in France to foreign parents at the age of 18 and that the obligation for citizens to undergo military service makes naturalization a costly choice. We exploit the abolition of compulsory military service in France as an exogenous shock that reduced the cost of naturalization for second-generation men while not impacting women.

We show that the decrease in the cost of naturalization led to a sudden increase in naturalization rates. In line with the theoretical framework developed to rationalize take-up decisions, this increase is entirely driven by individuals with the lowest expected benefits: individuals born European Union citizens who can already work and reside in France. They take up citizenship less often when it is tied to doing military service, while this cost is not binding for other birth nationality groups. Additionally, we show that there is a skill gradient in the cost of military service, with the low-educated reacting more to the reform. This feature is reminiscent of policies that are explicitly designed to

affect naturalization rates such as language tests or financial requirements.

The fact that EU males are the only group to react to the reform allows us to exploit the variety of unaffected gender and birth nationality groups in a Synthetic Difference-in-Differences approach to study the causal impact of this rise in naturalization rates on labor market outcomes. Our results show positive labor market effects, with an increase in employment and a reduction in inactivity for EU citizens. We find that this employment effect is accompanied by an increase in public sector employment and by a decrease in self-employment.

Altogether, these results provide valuable insights both on the citizenship take-up decision of children of immigrants, and on the gains from naturalization for those who would have renounced French citizenship had it entailed higher costs. In particular, our findings illustrate how policies that affect naturalization costs heterogeneously across immigrants can unintentionally divest specific groups of improved labor market prospects.

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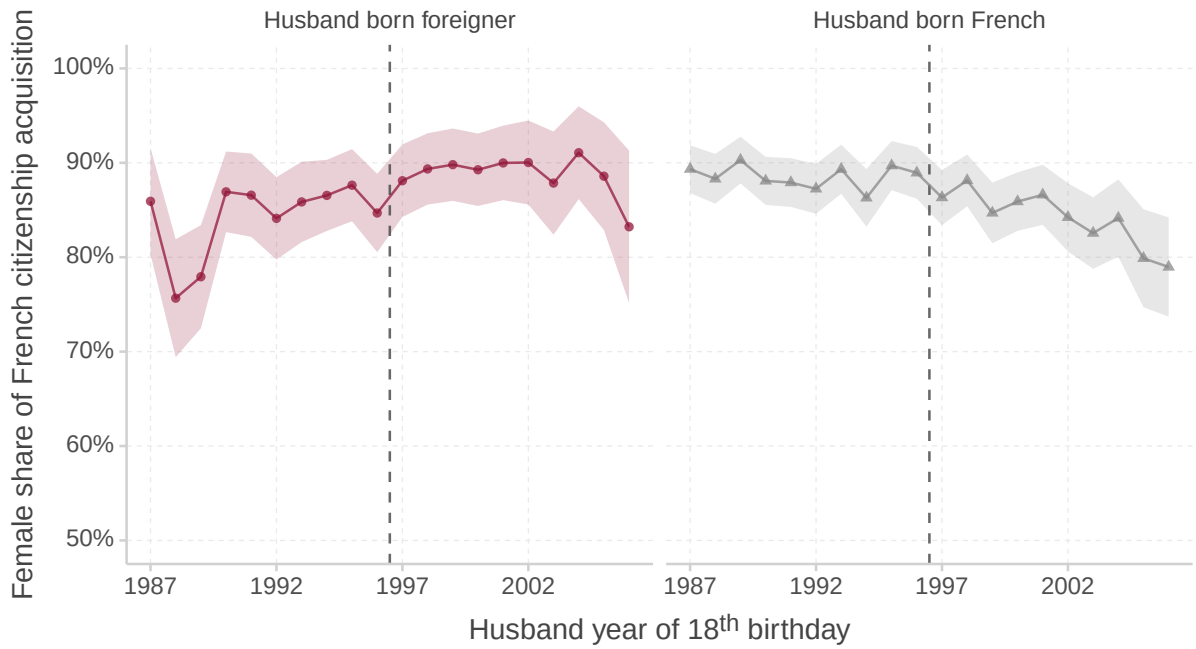
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Appendix

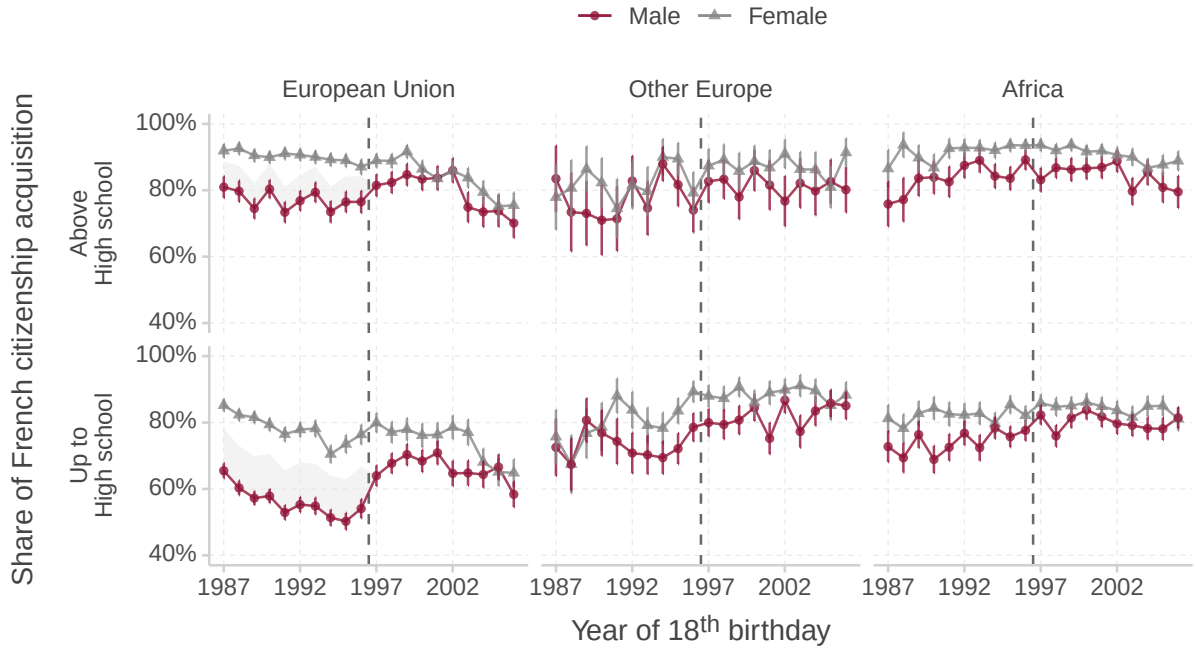
A Additional figure

Figure A.1: Female naturalization rates by their husband's birth cohort



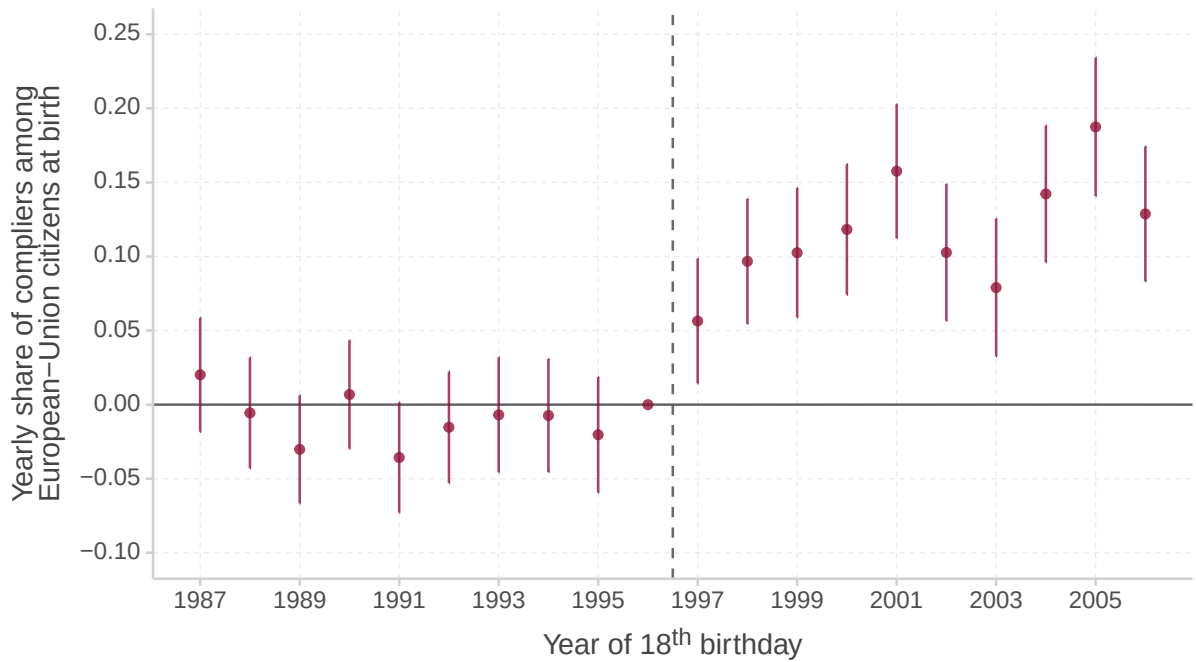
Notes: This figure represents the share of French citizenship acquisition among women born in France without French citizenship according to the birth cohort of their husband. The sample is restricted to women whose husband was born in France, and results are presented separately for those whose husband was born French (left panel) and those whose husband was born foreigner (right panel). The x -axis is labeled according to the husband's year of 18th birthday, which is when those born before 1979 must decide whether to do military service or to renounce French citizenship. The vertical dashed line represents the abolition of compulsory conscription. 95% confidence intervals are represented with light ribbons. Source: *French Population Census*, wave 2014 - complementary census sample, INSEE.

Figure A.2: Naturalization rates by education and birth nationality groups - Excluding Austrian, Finnish, and Swedish birth nationalities



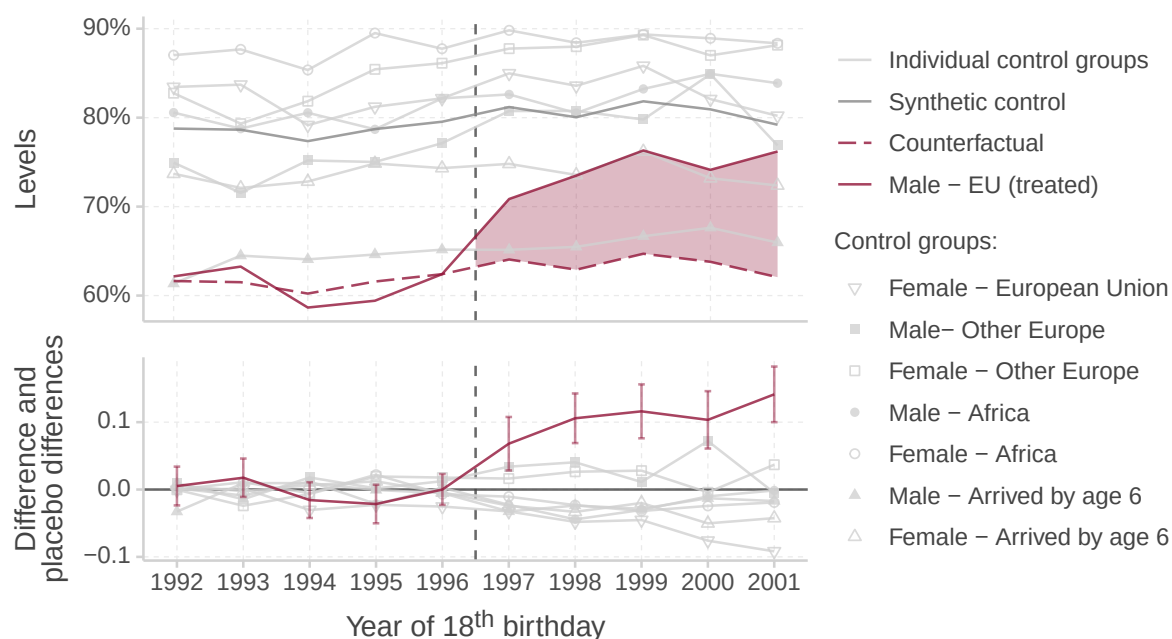
Notes: This figure represents the share of French citizenship acquisition among individuals born in France without French citizenship for 6 subgroups defined according to birth nationality (European Union, Other Europe, and Africa) and education (up to high school and above high school). European Union is defined as it was in 1994, before the inclusion of Austria, Finland, and Sweden. In each panel, shares of citizenship acquisition are represented separately for males (red) and females (gray), and for birth cohorts from 1969 to 1988. Vertical lines show the corresponding 95% confidence intervals. The x -axis is labeled according to the year of 18th birthday, which is when males born before 1979 must decide whether to do military service or to renounce French citizenship. Shaded areas represent the fraction of missing citizenship take-up among young males caused by compulsory conscription. The height of the area is obtained from Difference-in-Differences regressions between males and females born 5 years before and after 1979, estimated on the corresponding subgroups as specified in Equation 1. Regression results are reported in Appendix Table B.3. A significant effects are found only for individuals born citizens of the European Union. Source: *French Population Census*, wave 2014, INSEE.

Figure A.3: Naturalization rates of EU second-generation immigrants



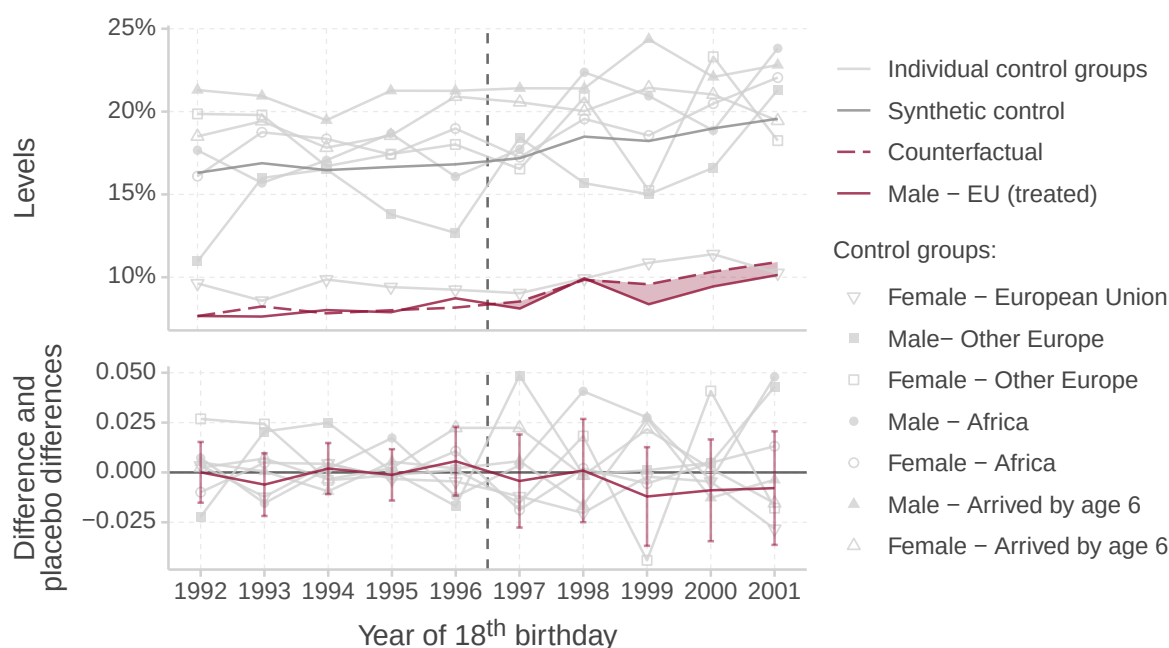
Notes: This figure shows regression results which document the evolution of the difference in French citizenship acquisition rates between males and females among those born in France with a non-French European Union citizenship. European Union is defined as it was in 1996. Coefficients are obtained from the regression of French citizenship acquisition on gender, birth cohort, and their interaction. The 1978 birth cohort is used as the reference category because it is the last cohort which was not subject to compulsory conscription. Females are used as the reference category because they were not subject to the abolition of compulsory conscription. Source: *French Population Census*, wave 2014, INSEE.

Figure A.4: Synthetic Difference-in-Differences - Naturalization



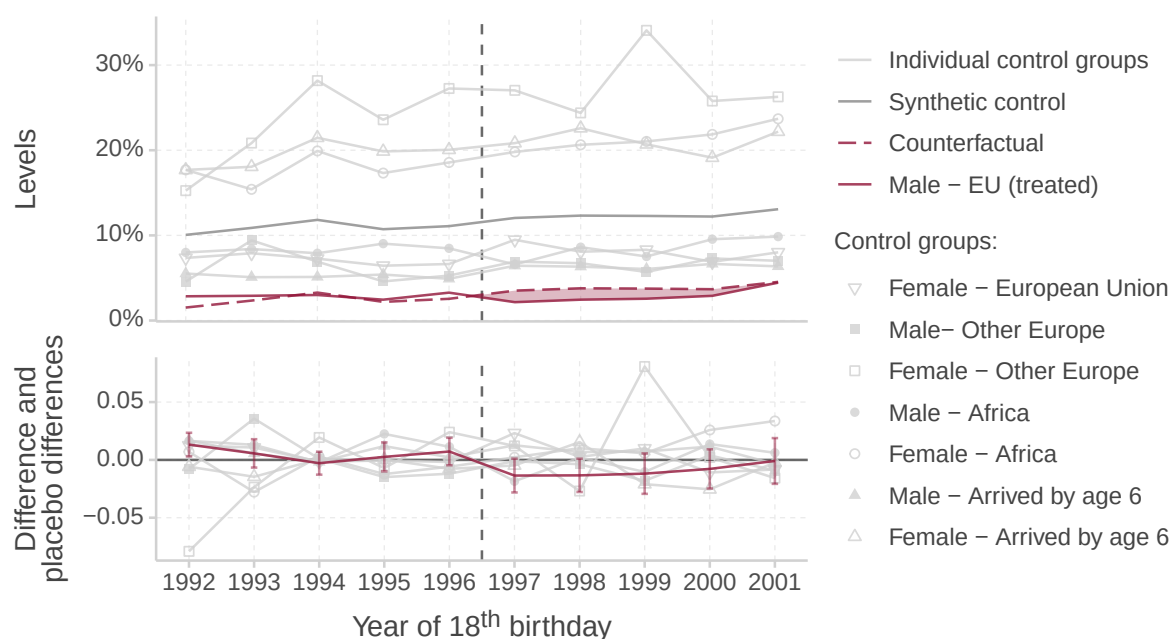
Notes: The top panel of this figure shows the share of individuals who acquired French citizenship by birth cohort within each of the groups used in our Synthetic Difference-in-Differences identification strategy. That of the treated group, male European Union citizens, is represented as a solid red line. That of each individual control group is represented with a gray line, whose markers indicate the corresponding group. The solid gray line shows the trend of the synthetic control group. The dashed red line represents the counterfactual trend of the treated group. It corresponds to the trend of the synthetic control group shifted by the average difference between the trend of the treatment group and that of the synthetic control group in the pre-period. The bottom panel displays the difference between the treated group and the synthetic control group with a solid red line, centered at the pre-treatment average difference weighted by the estimated time weights. Vertical lines show the corresponding 95% bootstrapped confidence intervals. The gray lines are placebo effects, each using one of the control groups as the treatment group instead of EU males. Corresponding regression results are reported in Table B.8. Source: *French Population Census*, wave 2014, INSEE.

Figure A.5: Synthetic Difference-in-Differences - Unemployed



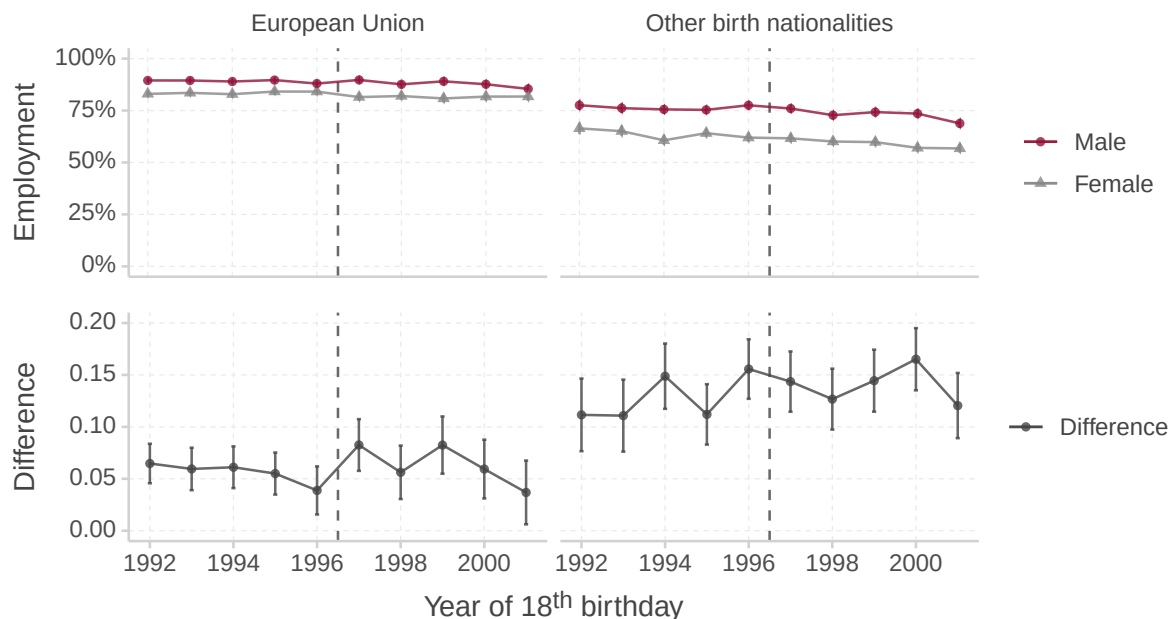
Notes: The top panel of this figure shows the share of unemployed individuals by birth cohort within each of the groups used in our Synthetic Difference-in-Differences identification strategy. That of the treated group, male European Union citizens, is represented as a solid red line. That of each individual control group is represented with a gray line, whose markers indicate the corresponding group. The solid gray line shows the trend of the synthetic control group. The dashed red line represents the counterfactual trend of the treated group. It corresponds to the trend of the synthetic control group shifted by the average difference between the trend of the treatment group and that of the synthetic control group in the pre-period. The bottom panel displays the difference between the treated group and the synthetic control group with a solid red line, centered at the pre-treatment average difference weighted by the estimated time weights. Vertical lines show the corresponding 95% bootstrapped confidence intervals. The gray lines are placebo effects, each using one of the control groups as the treatment group instead of EU males. Corresponding regression results are reported in Table B.8. Source: *French Population Census*, wave 2014, INSEE.

Figure A.6: Synthetic Difference-in-Differences - Inactive



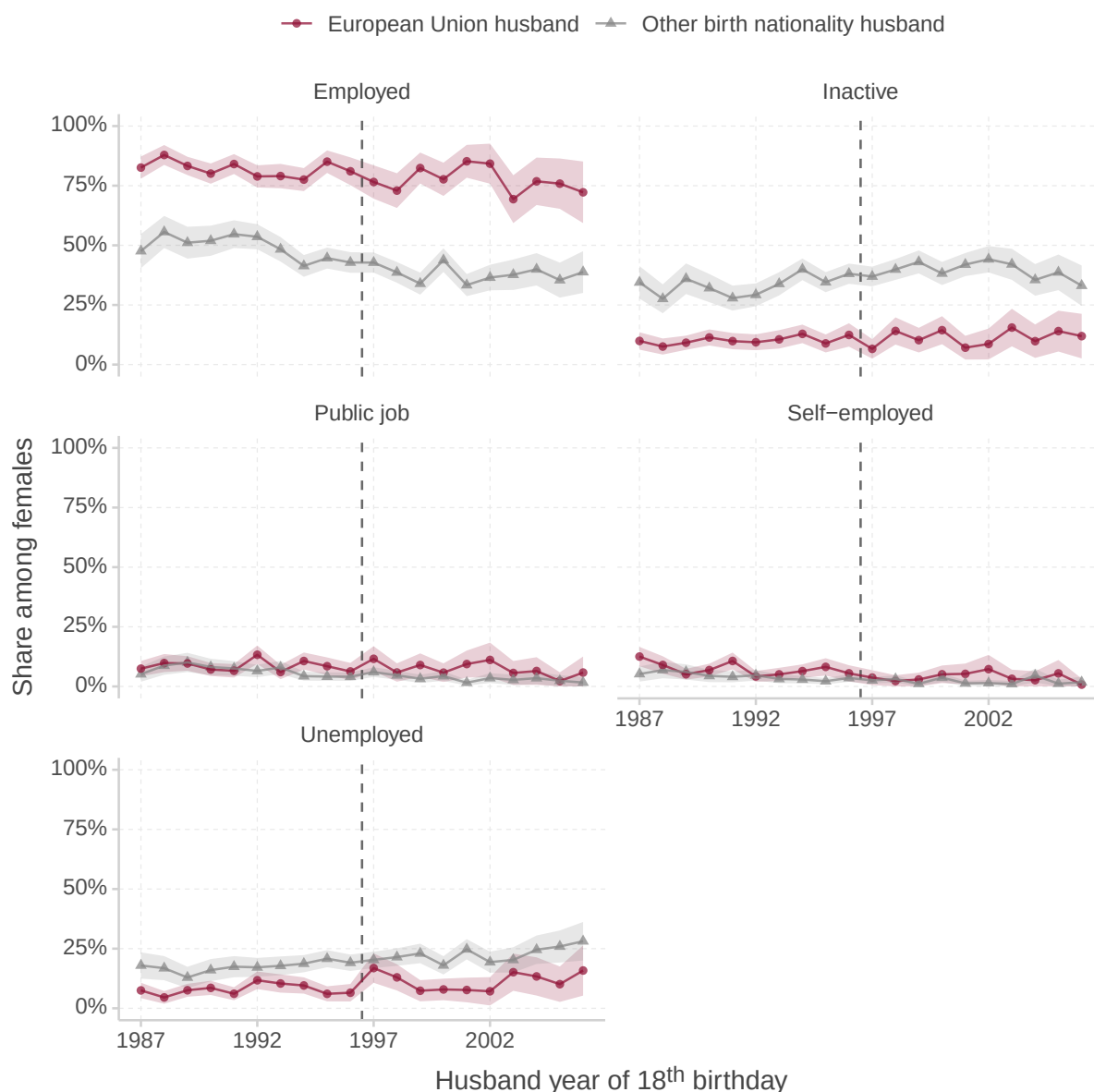
Notes: The top panel of this figure shows the share of inactive individuals by birth cohort within each of the groups used in our Synthetic Difference-in-Differences identification strategy. That of the treated group, male European Union citizens, is represented as a solid red line. That of each individual control group is represented with a gray line, whose markers indicate the corresponding group. The solid gray line shows the trend of the synthetic control group. The dashed red line represents the counterfactual trend of the treated group. It corresponds to the trend of the synthetic control group shifted by the average difference between the trend of the treatment group and that of the synthetic control group in the pre-period. The bottom panel displays the difference between the treated group and the synthetic control group with a solid red line, centered at the pre-treatment average difference weighted by the estimated time weights. Vertical lines show the corresponding 95% bootstrapped confidence intervals. The gray lines are placebo effects, each using one of the control groups as the treatment group instead of EU males. Corresponding regression results are reported in Table B.8. Source: *French Population Census*, wave 2014, INSEE.

Figure A.7: Difference in employment trends between males and females - EU vs. Non-EU citizens at birth



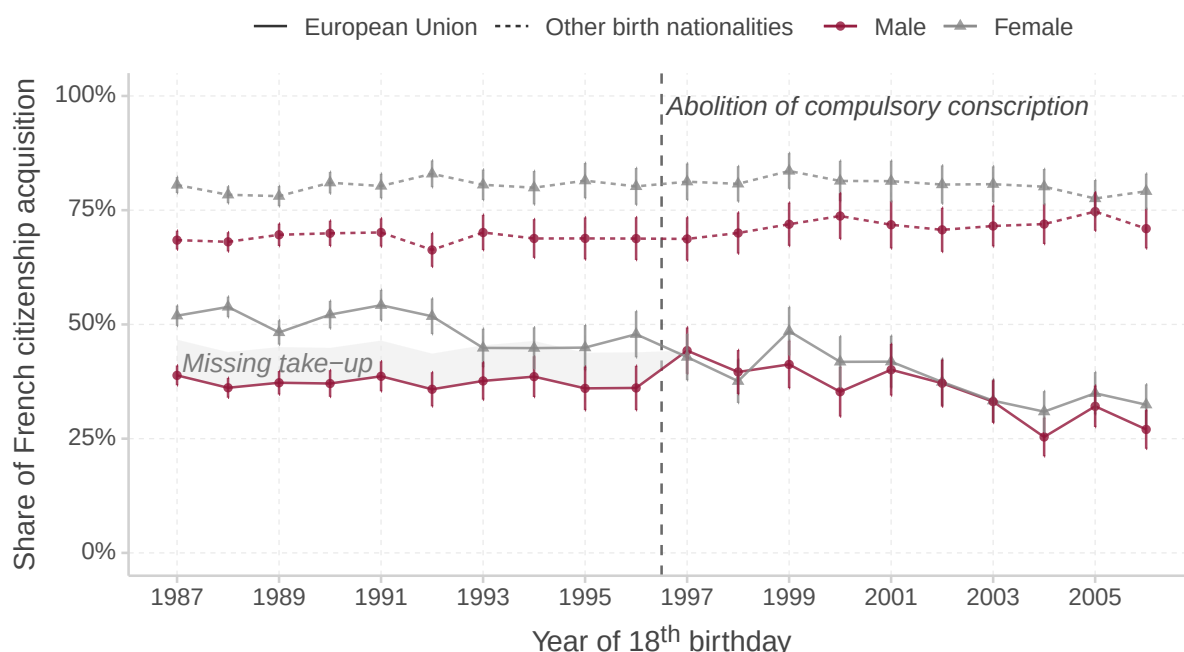
Notes: The top panel of this figure shows the share of employed males (red) and females (gray) by birth cohort among individuals born in France without French citizenship, separately for those born citizens of the European Union and those born with a nationality outside the EU. European Union is defined as it was in 1996. The x -axis is labeled according to the year of 18th birthday, which is when men born before 1979 must decide whether to do military service or to renounce French citizenship. The vertical dashed line represents the abolition of compulsory conscription. 95% confidence intervals are represented with vertical error bars. The bottom panel shows the difference in the share of employed individuals between males and females, by birth cohort and group of birth nationality. Source: *French Population Census*, wave 2014, INSEE.

Figure A.8: Female labor market outcomes by their husband's birth cohort



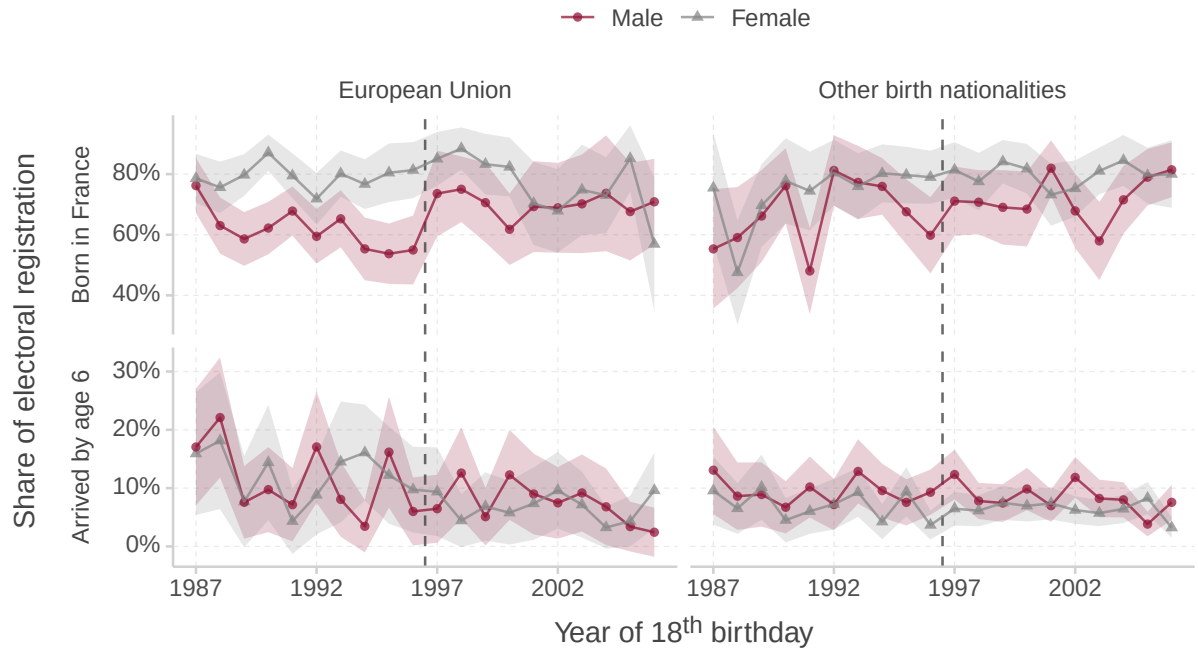
Notes: This figure represents average labor market outcomes of women born in France without French citizenship according to the birth cohort of their husband. The sample is restricted to women whose husband was born in France without French citizenship. Each panel corresponds to a given labor market outcome, and represents the share of women which pertains to the corresponding category, separately for those whose husband's birth citizenship pertains to the European Union (red) and for others (gray). European Union is defined as it was in 1996. The x-axis is labeled according to the husband's year of 18th birthday, which is when those born before 1979 must decide whether to do military service or to renounce French citizenship. The vertical dashed line represents the abolition of compulsory conscription. 95% confidence intervals are represented with light ribbons, bounded between 0 and 1. Source: *French Population Census*, wave 2014 - complementary census sample, INSEE.

Figure A.9: Naturalization rates of first-generation immigrants arrived by age 6



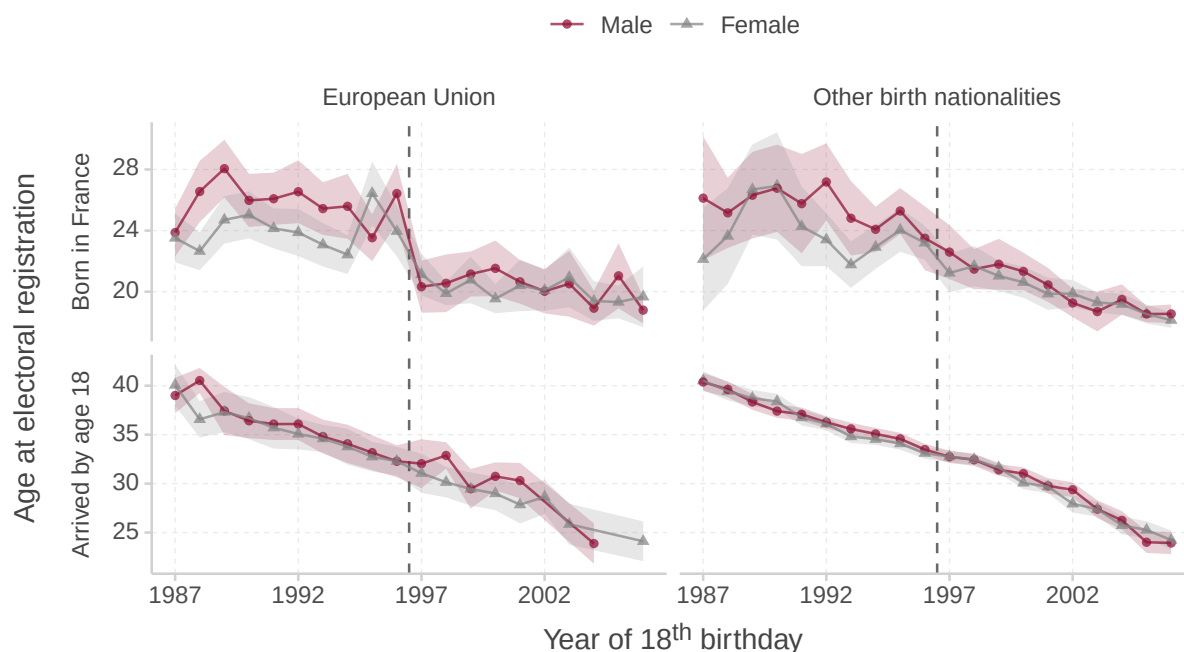
Notes: This figure represents the share of French citizenship acquisition among individuals born abroad and arrived in France by age 6 without French citizenship, separately for males (red) and females (gray), for those born with an European Union citizenship (solid lines) and other citizenship (dashed lines), for birth cohorts from 1969 to 1988. European Union is defined as it was in 1996. Vertical lines show the corresponding 95% confidence intervals. The x -axis is labeled according to the year of 18th birthday, when military service is tied to French citizenship acquisition for men. The vertical dashed line represents the abolition of compulsory conscription. For males, after this point, citizenship acquisition is not tied to doing military service anymore. The shaded area represents the estimated fraction of missing citizenship take-up caused by compulsory conscription among young males born citizens of the European Union without French citizenship, who were born abroad and who arrived in France by age 6. The height of the area is obtained from a Difference-in-Differences regression between males and females born 5 years before and after 1979. Source: *French Population Census*, wave 2014, INSEE.

Figure A.10: Registration in electoral list



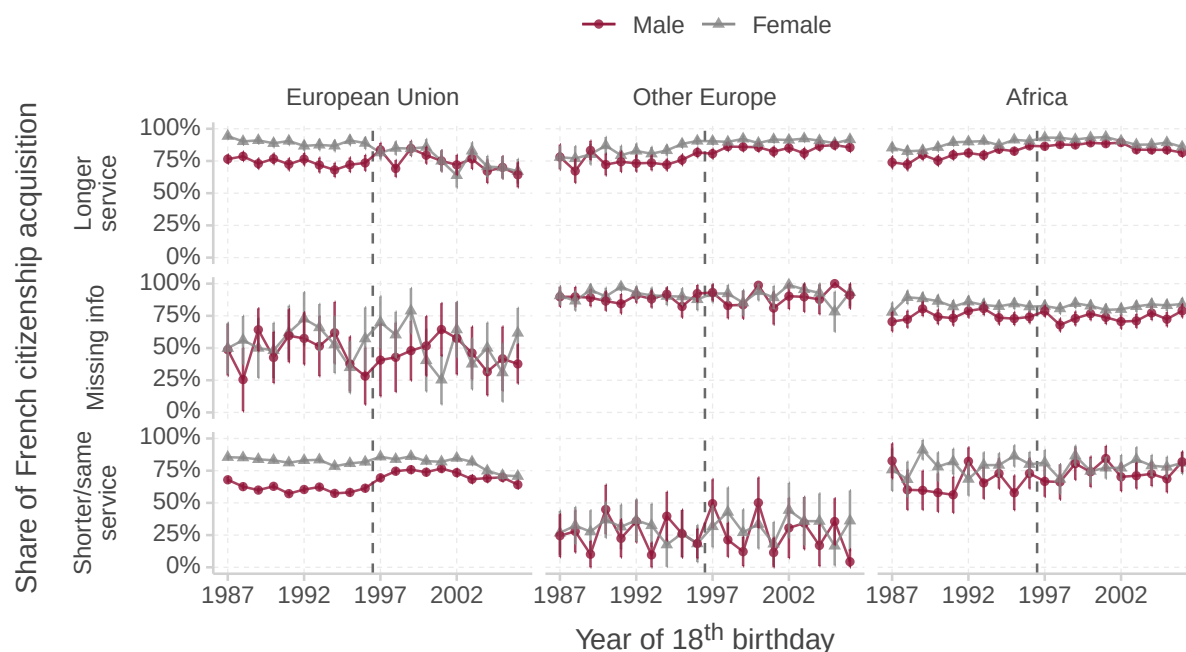
Notes: This figure shows the share of individuals registered in the electoral registers, as a proxy for citizenship acquisition, for 4 different subgroups defined according to place of birth (born France vs. born abroad and arrived in France by age 6) and group of birth citizenship (European Union citizenship vs. others). European Union defined as it was in 1996. In each panel, registration rates are represented separately for males (red) and females (gray), for each birth cohort from 1969 to 1988. The x -axis is labeled according to the year of 18th birthday, which is when males born before 1979 must decide whether to do military service or to renounce French citizenship. 95% confidence intervals are represented with light ribbons. Computations are made on the Permanent Demographic Sample, which covers about 4% of the population. Source: *Permanent Demographic Sample*, INSEE-DGFIP.

Figure A.11: Age at registration in electoral list



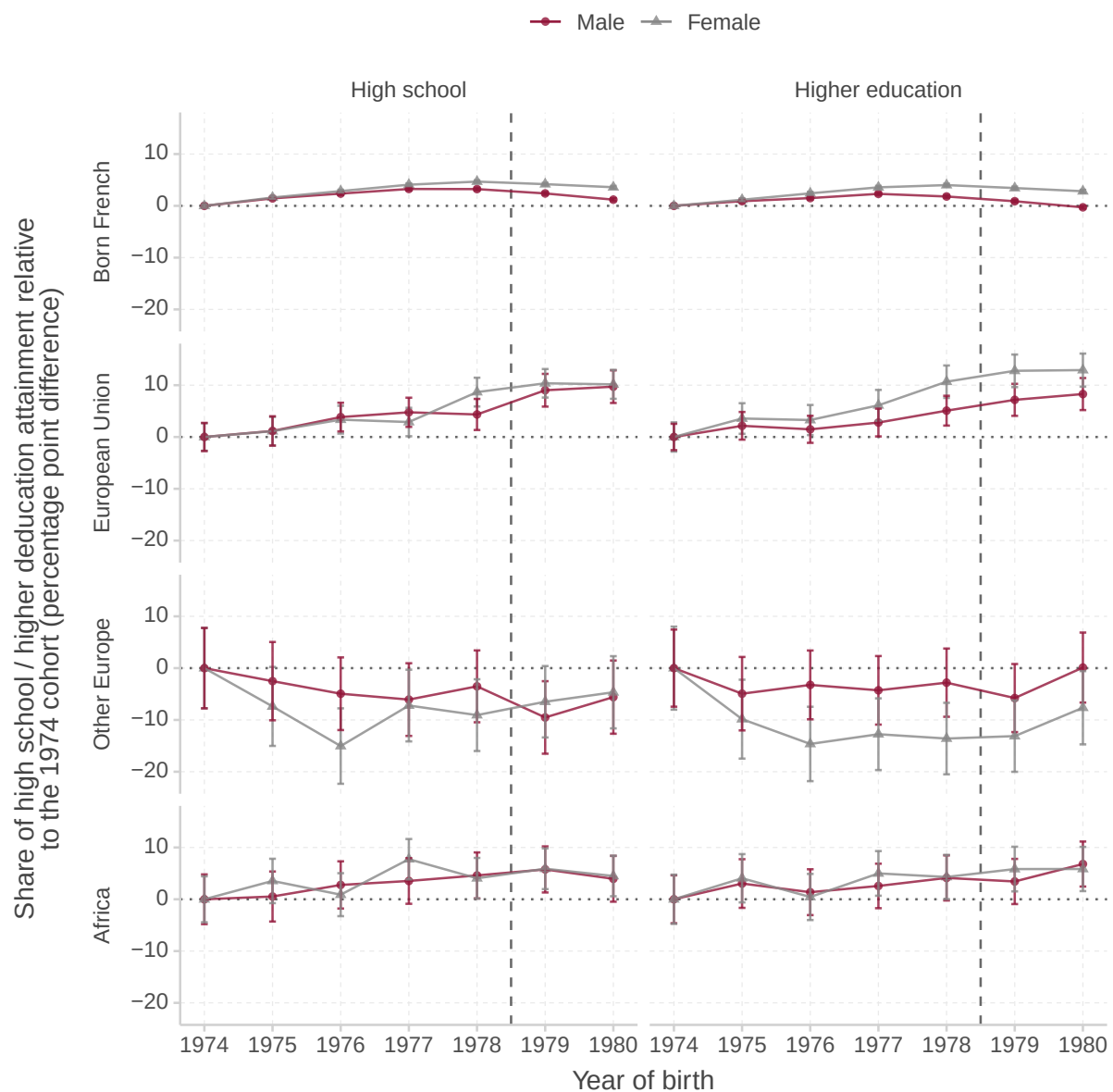
Notes: This figure shows the average age at registration in the electoral registers, as a proxy for age at acquisition of citizenship, for 4 different subgroups defined according to place of birth (born France vs. born abroad and arrived in France by age 18) and group of birth citizenship (European Union citizenship vs. others). European Union defined as it was in 1996. In each panel, registration rates are represented separately for males (red) and females (gray), for each birth cohort from 1979 to 1988. The x -axis is labeled according to the year of 18th birthday, which is when males born before 1979 must decide whether to do military service or to renounce French citizenship. 95% confidence intervals are represented with light ribbons. Computations are made on the Permanent Demographic Sample, which covers about 4% of the population. Missing information for some groups in specific years is due to insufficient sample size. Source: *Permanent Demographic Sample*, INSEE-DGFIP.

Figure A.12: Naturalization rates by duration of the military service of birth nationality



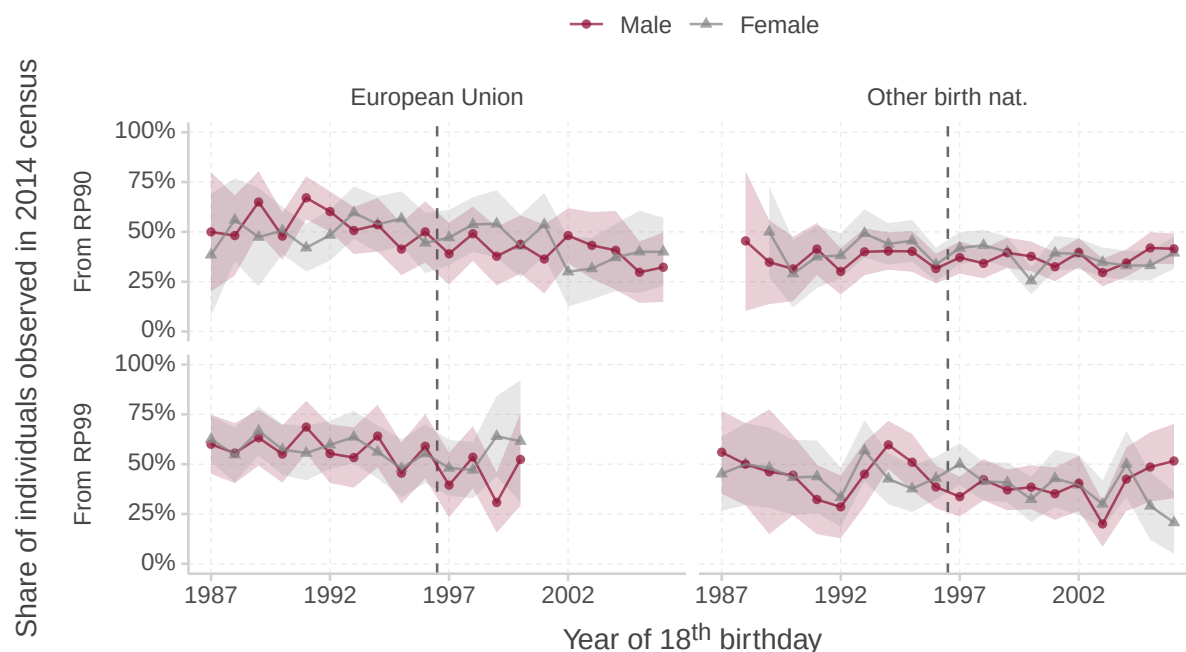
Notes: This figure represents the share of French citizenship acquisition among individuals born in France without French citizenship for 9 subgroups defined by birth nationality (European Union, Other Europe, and Africa) and military service duration of in the country of birth nationality (longer vs. shorter than the French military service, or unknown). European Union is defined as it was in 1996. In each panel, shares of citizenship acquisition are represented separately for males (red) and females (gray), for birth cohorts from 1969 to 1988. Vertical lines show the corresponding 95% confidence intervals, bounded between 0 and 1. The x -axis is labeled according to the year of 18th birthday, which is when males born before 1979 must decide whether to do military service or to renounce French citizenship. Source: *French Population Census*, wave 2014, INSEE.

Figure A.13: Education rates relative to the 1974 birth cohort



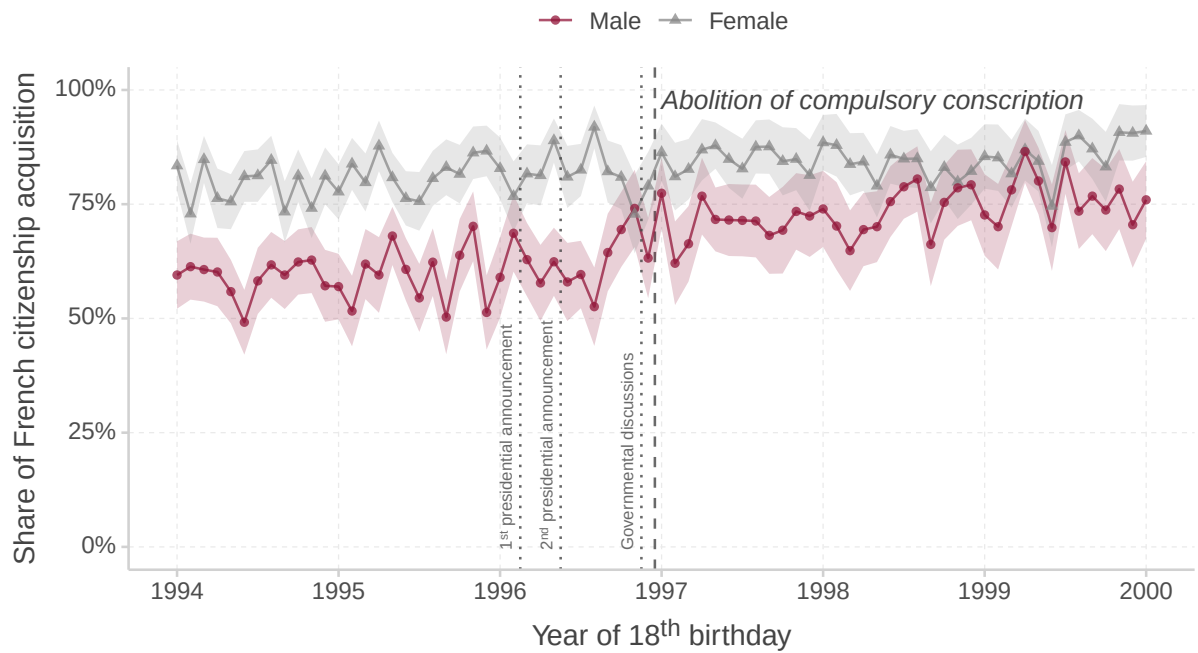
Notes: This figure represents the share of educational attainment -high school on the left panel and higher education on the right panel- for individuals born in France separately by gender, birth cohort, and for four birth nationality groups: French nationality, European Union nationalities, other European nationalities, and African nationalities. European Union is defined as it was in 1996. Each point represents the difference between the share of a given educational attainment in the corresponding birth cohort and in the 1974 birth cohort, following [Maurin and Xenogiani \(2007\)](#). Vertical bars represent the 95% confidence intervals, and the vertical dashed lines represent the moment of the abolition of compulsory conscription. Source: *French Population Census*, wave 2014, INSEE.

Figure A.14: Rates of presence in the 2014 census among 1990 and 1999 census populations



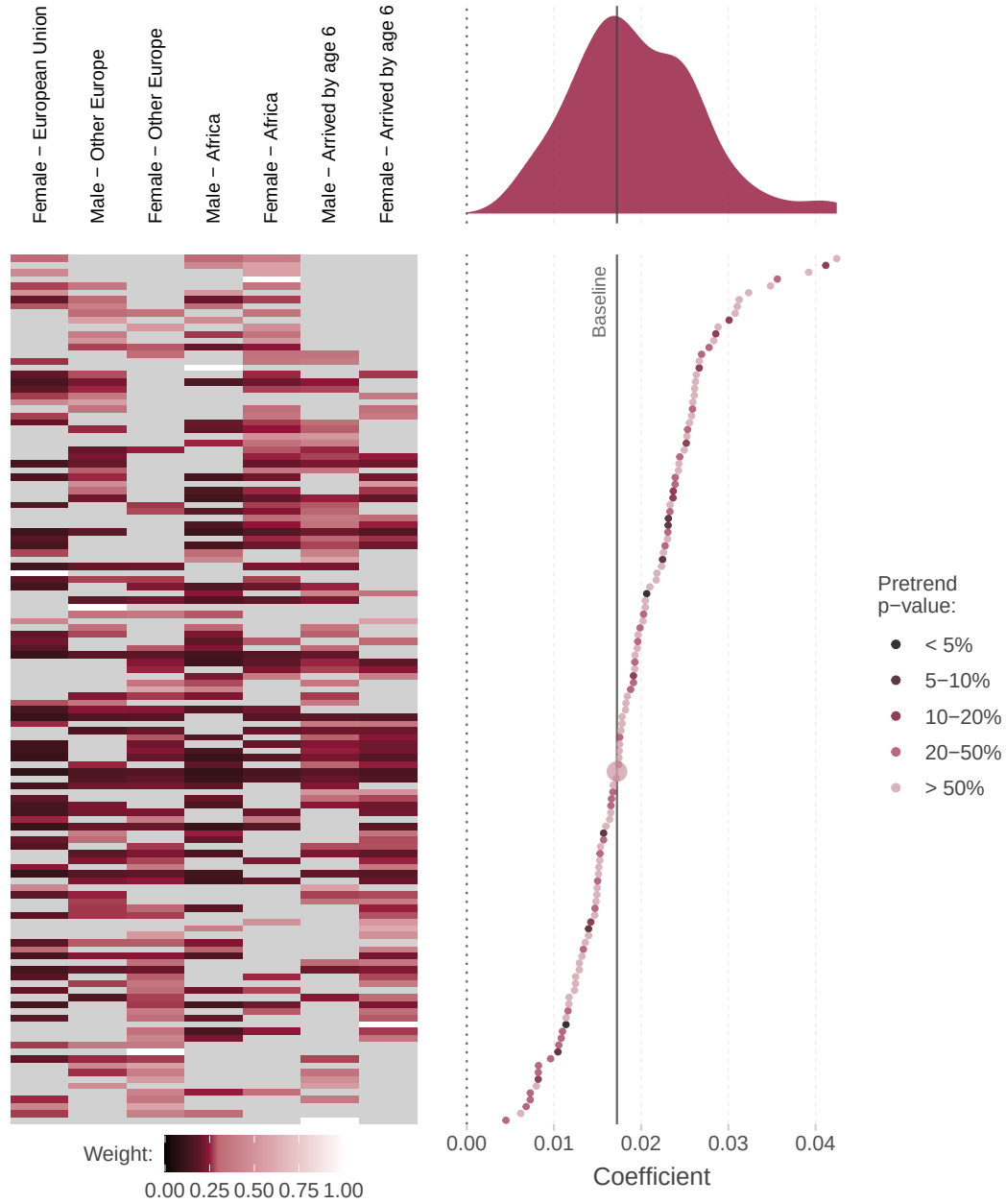
Notes: This figure represents the share of individual observed in the 2014 census wave among individuals observed in the 1990 census wave (top panel) and in the 1999 census wave (lower panel), separately for males (red) and females (gray) and for birth cohorts from 1969 to 1988. The x -axis is labeled according to the year of 18th birthday, which is when males born before 1979 must decide whether to do military service or to refuse French citizenship. The vertical dashed line represents the abolition of compulsory conscription. Computations are made on the Permanent Demographic Sample, in which census information for the 1990 and 1999 waves is available for individuals born during the first 4 days of October. Missing information for some groups in specific years is due to insufficient sample size. Source: *Permanent Demographic Sample*, INSEE-DGFIP.

Figure A.15: Anticipation of the reform - Naturalization rates by month of birth



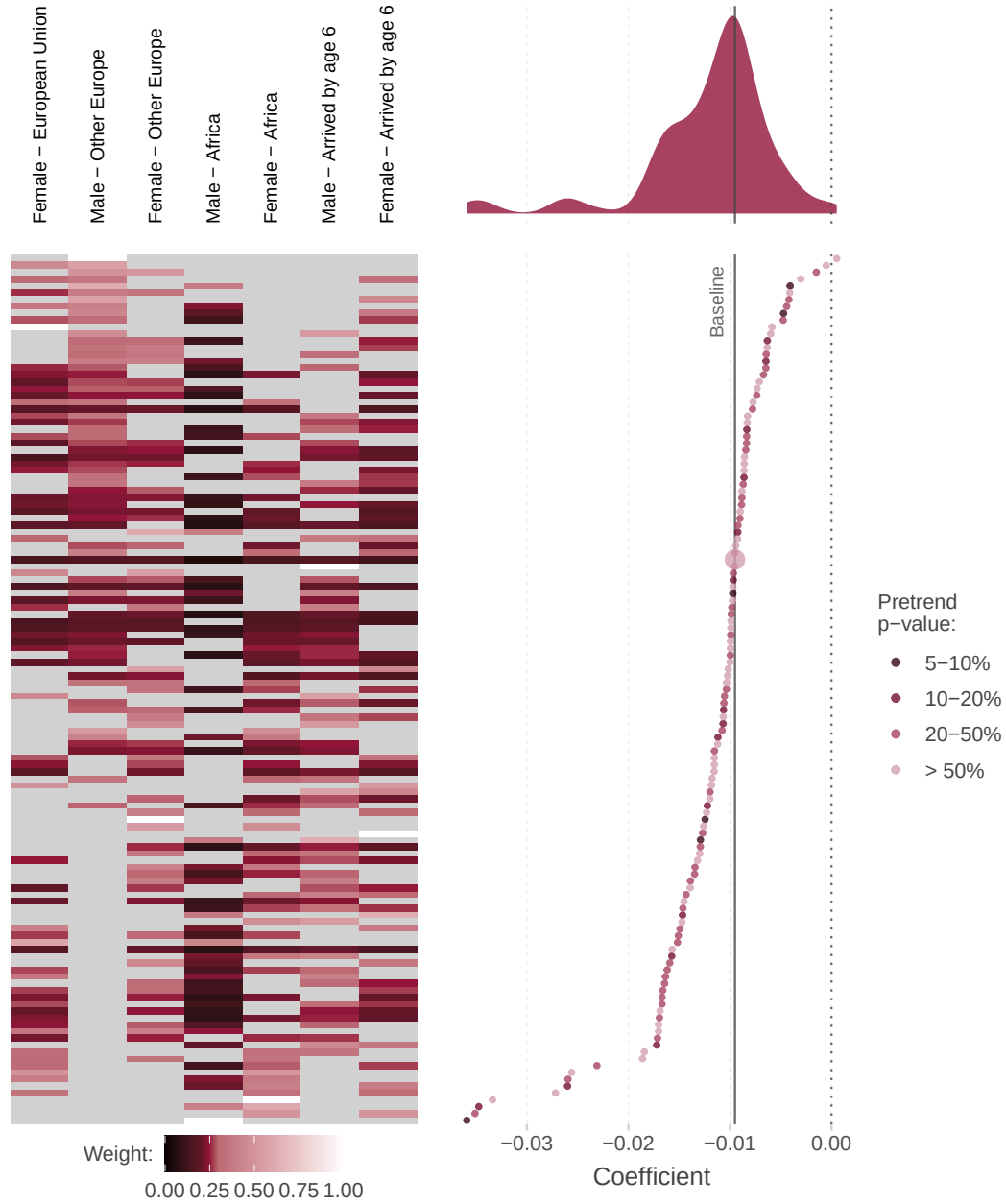
Notes: This figure represents the share of French citizenship acquisition among individuals born in France without French citizenship, separately for males (red) and females (gray), for each month of birth from January 1976 to January 1982. The x -axis is labeled according to the month of 18th birthday. 95% confidence intervals are represented with light ribbons. The vertical dashed line represents the abolition of compulsory conscription. For males, after this point, citizenship acquisition is not tied to doing military service anymore. The vertical dotted lines represent milestones in the implementation of the reform, from the 1st presidential announcement to the final governmental discussions. Source: *French Population Census*, wave 2014, INSEE.

Figure A.16: Employment ITT - Every combination of control groups



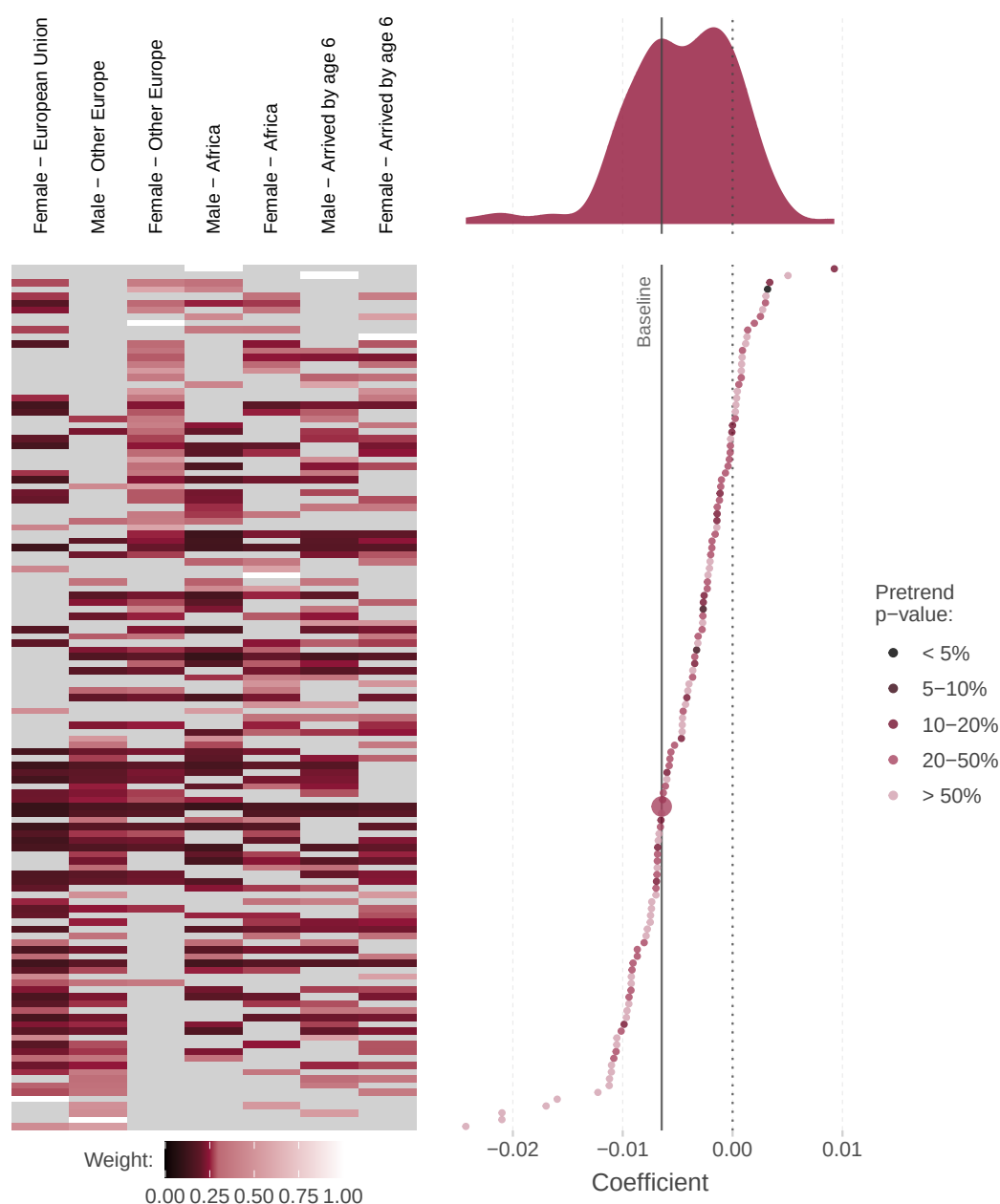
Notes: This figure shows the results of Synthetic Difference-in-Differences estimations of the effect of the increase in the naturalization rates among EU males, induced by the abolition of compulsory conscription, on the share of employed individuals. Specifically, it shows the ITT effect on the share of employed individuals estimated for each of the 127 possible combinations formed by sets of 1 to 7 control groups to generate the synthetic control group. Each row of the tile plot corresponds to a given estimation. A tile whose color lies on the gradient in the legend indicates that the control group of the corresponding column was included in the computation of the synthetic control group, while a gray tile indicates that it has been excluded. The specification which includes the whole set of 7 control groups is our baseline specification. Specifications are sorted by the value of the corresponding ITT coefficient reported on the scatter plot on the right panel. The color of each dot indicates the p-value of the pretrend. To obtain this value, we compute the difference between the average outcome in the treated group and that in the synthetic control group, and regress it on the birth cohort for the years before the reform. The p-value is then computed from a standard two-sided t-test on the slope coefficient. The vertical solid line represents our baseline coefficient, and a vertical dashed line is placed at 0. The kernel density of this distribution of 127 coefficients is shown on the top panel. Source: *French Population Census*, wave 2014, INSEE.

Figure A.17: Inactive ITT - Every combination of control groups



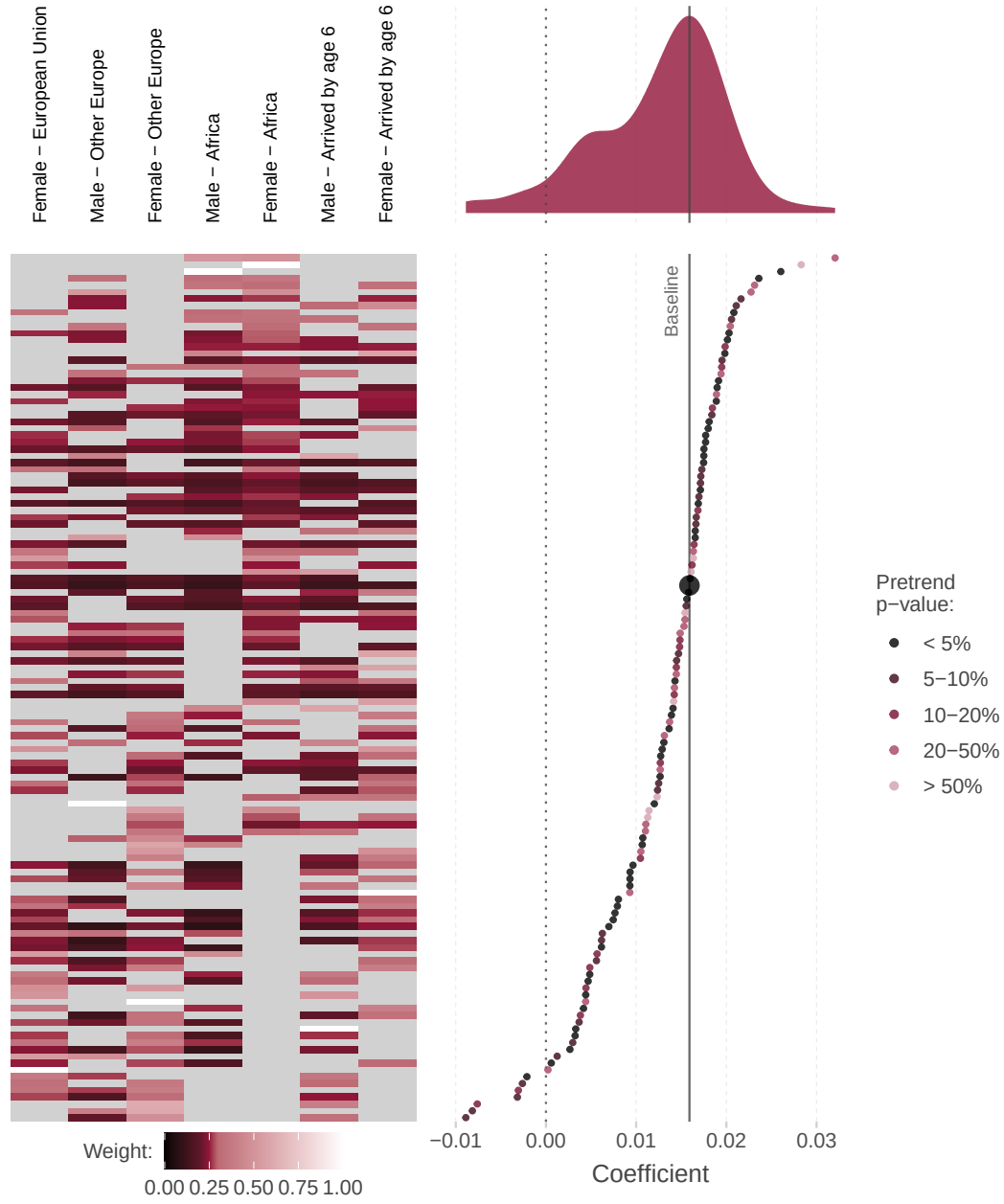
Notes: This figure shows the results of Synthetic Difference-in-Differences estimations of the effect of the increase in the naturalization rates among EU males, induced by the abolition of compulsory conscription, on the share of inactive individuals. Specifically, it shows the ITT effect on the share of inactive individuals estimated for each of the 127 possible combinations formed by sets of 1 to 7 control groups to generate the synthetic control group. Each row of the tile plot corresponds to a given estimation. A tile whose color lies on the gradient in the legend indicates that the control group of the corresponding column was included in the computation of the synthetic control group, while a gray tile indicates that it has been excluded. The specification which includes the whole set of 7 control groups is our baseline specification. Specifications are sorted by the value of the corresponding ITT coefficient reported on the scatter plot on the right panel. The color of each dot indicates the p-value of the pre-trend. To obtain this value, we compute the difference between the average outcome in the treated group and that in the synthetic control group, and regress it on the birth cohort for the years before the reform. The p-value is then computed from a standard two-sided t-test on the slope coefficient. The vertical solid line represents our baseline coefficient, and a vertical dashed line is placed at 0. The kernel density of this distribution of 127 coefficients is shown on the top panel. Source: *French Population Census*, wave 2014, INSEE.

Figure A.18: Unemployed ITT - Every combination of control groups



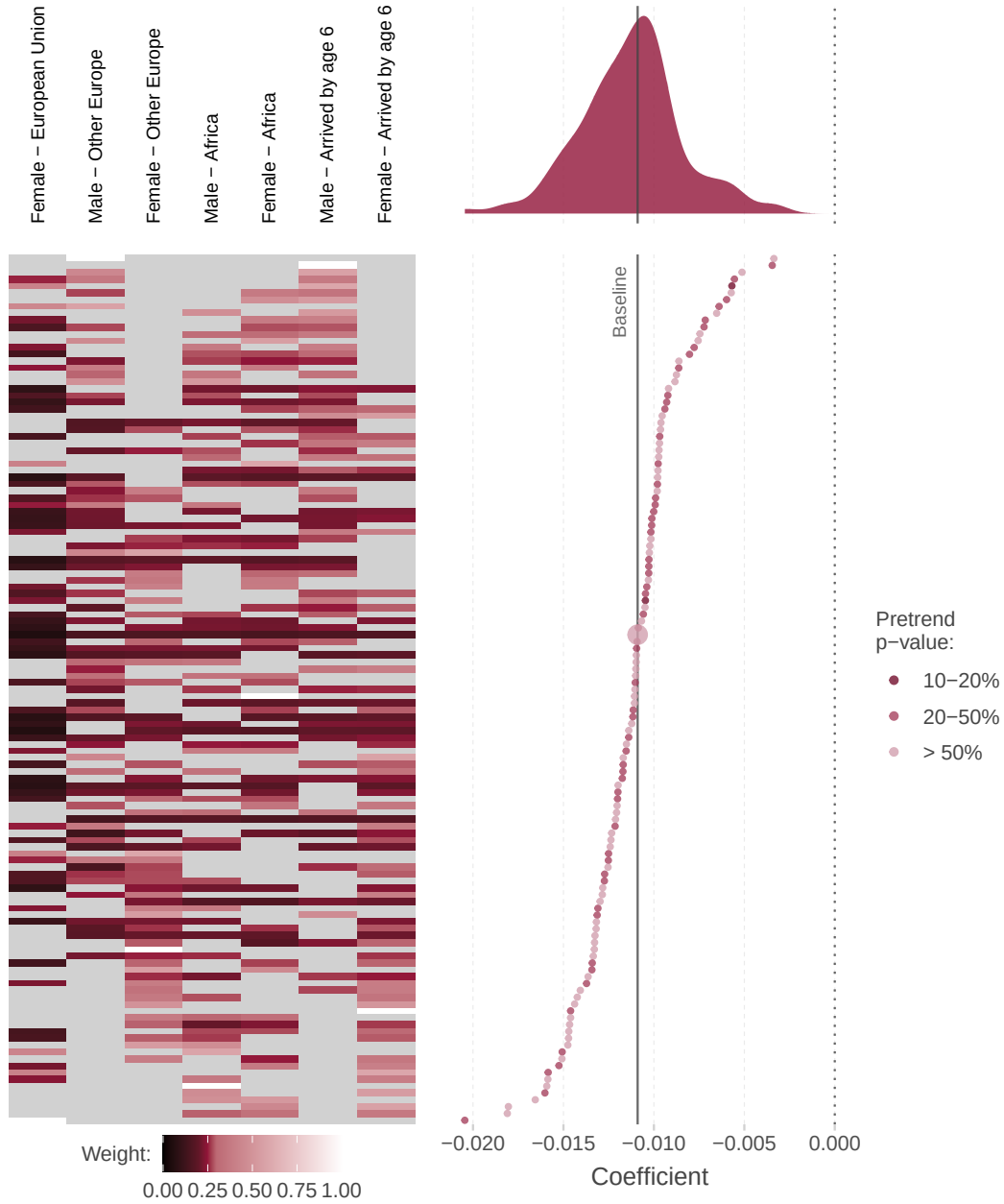
Notes: This figure shows the results of Synthetic Difference-in-Differences estimations of the effect of the increase in the naturalization rates among EU males, induced by the abolition of compulsory conscription, on the share of unemployed individuals. Specifically, it shows the ITT effect on the share of unemployed individuals estimated for each of the 127 possible combinations formed by sets of 1 to 7 control groups to generate the synthetic control group. Each row of the tile plot corresponds to a given estimation. A tile whose color lies on the gradient in the legend indicates that the control group of the corresponding column was included in the computation of the synthetic control group, while a gray tile indicates that it has been excluded. The specification which includes the whole set of 7 control groups is our baseline specification. Specifications are sorted by the value of the corresponding ITT coefficient reported on the scatter plot on the right panel. The color of each dot indicates the p-value of the pretrend. To obtain this value, we compute the difference between the average outcome in the treated group and that in the synthetic control group, and regress it on the birth cohort for the years before the reform. The p-value is then computed from a standard two-sided t-test on the slope coefficient. The vertical solid line represents our baseline coefficient, and a vertical dashed line is placed at 0. The kernel density of this distribution of 127 coefficients is shown on the top panel. Source: *French Population Census*, wave 2014, INSEE.

Figure A.19: Public job ITT - Every combination of control groups



Notes: This figure shows the results of Synthetic Difference-in-Differences estimations of the effect of the increase in the naturalization rates among EU males, induced by the abolition of compulsory conscription, on public-sector employment. Specifically, it shows the ITT effect on the share of public-sector employment estimated for each of the 127 possible combinations formed by sets of 1 to 7 control groups to generate the synthetic control group. Each row of the tile plot corresponds to a given estimation. A tile whose color lies on the gradient in the legend indicates that the control group of the corresponding column was included in the computation of the synthetic control group, while a gray tile indicates that it has been excluded. The specification which includes the whole set of 7 control groups is our baseline specification. Specifications are sorted by the value of the corresponding ITT coefficient reported on the scatter plot on the right panel. The color of each dot indicates the p-value of the pre-trend. To obtain this value, we compute the difference between the average outcome in the treated group and that in the synthetic control group, and regress it on the birth cohort for the years before the reform. The p-value is then computed from a standard two-sided t-test on the slope coefficient. The vertical solid line represents our baseline coefficient, and a vertical dashed line is placed at 0. The kernel density of this distribution of 127 coefficients is shown on the top panel. Source: *French Population Census*, wave 2014, INSEE.

Figure A.20: Self-employed ITT - Every combination of control groups



Notes: This figure shows the results of Synthetic Difference-in-Differences estimations of the effect of the increase in the naturalization rates among EU males, induced by the abolition of compulsory conscription, on self-employment. Specifically, it shows the ITT effect on the share of self-employment estimated for each of the 127 possible combinations formed by sets of 1 to 7 control groups to generate the synthetic control group. Each row of the tile plot corresponds to a given estimation. A tile whose color lies on the gradient in the legend indicates that the control group of the corresponding column was included in the computation of the synthetic control group, while a gray tile indicates that it has been excluded. The specification which includes the whole set of 7 control groups is our baseline specification. Specifications are sorted by the value of the corresponding ITT coefficient reported on the scatter plot on the right panel. The color of each dot indicates the p-value of the pre-trend. To obtain this value, we compute the difference between the average outcome in the treated group and that in the synthetic control group, and regress it on the birth cohort for the years before the reform. The p-value is then computed from a standard two-sided t-test on the slope coefficient. The vertical solid line represents our baseline coefficient, and a vertical dashed line is placed at 0. The kernel density of this distribution of 127 coefficients is shown on the top panel. Source: *French Population Census*, wave 2014, INSEE.

B Additional tables

Table B.1: Share of the most represented birth nationalities of individuals born in France without French citizenship

European Union		Other Europe		Africa		Other	
Portugal	33.69%	Turkey	7.05%	Morocco	15.97%	Cambodia	0.84%
Spain	6.34%	Serbia	2.01%	Tunisia	7.61%	Lao	0.72%
Italy	6.34%	Poland	0.72%	Algeria	4.07%	Vietnam	0.53%
Belgium	0.72%	Switzerland	0.66%	Cameroon	0.85%	Haiti	0.52%
UK	0.68%	Romania	0.32%	Congo	0.70%	China	0.51%
Other	1.29%	Other	1.03%	Other	3.89%	Other	2.92%
Total	49.06%	Total	11.80%	Total	33.11%	Total	6.04%

Notes: This table reports the birth nationality composition of our sample: individuals born in France without French citizenship from 1969 to 1988. Note that Algerians are under-represented in our sample because most of them acquired French citizenship at birth under the principle of the “double jus soli”. Following the independence of Algeria, this principle applies to individuals born after January 1st 1963 to a parent born in Algeria before July 3rd 1962. European Union is defined as it was in 1996. Source: *French Population Census*, wave 2014, INSEE.

Table B.2: Effect of the abolition of compulsory conscription on naturalization

	French citizenship acquisition
Male	−0.152*** (0.004)
Post	0.027*** (0.004)
Post × Male	0.077*** (0.006)
Constant	0.837*** (0.003)
Observations	70,117
Mean dep. var.	0.794
R ²	0.029

Notes: This table shows the results of a Difference-in-Differences regression which captures the effect of the abolition of compulsory conscription on French citizenship acquisition. It is estimated on individuals born in France without French citizenship between 1974 and 1983. The dependent variable, French citizenship acquisition, is a dummy variable taking the value 1 if the individual acquired French citizenship and 0 otherwise. The variable Male is a dummy variable taking the value 1 for males and 0 for females. The variable Post is a dummy variable taking the value 1 for individuals born after 1978, i.e., for cohorts not subject to compulsory conscription, and 0 otherwise. Hence, the interaction Post × Male is also a dummy variable taking the value 1 for males born after 1978 and 0 otherwise. Standard errors are reported in parentheses, and significance level is reported according to the following symbology. *p<0.1; **p<0.05; ***p<0.01. Source: *French Population Census*, wave 2014, INSEE.

Table B.3: Effect of the abolition of compulsory conscription on naturalization by education and birth nationality group

	French citizenship acquisition					
	European Union		Other Europe		Africa	
	Low edu.	High edu.	Low edu.	High edu.	Low edu.	High edu.
Male	−0.222*** (0.008)	−0.128*** (0.008)	−0.107*** (0.014)	−0.036* (0.020)	−0.062*** (0.010)	−0.065*** (0.009)
Post	0.022** (0.010)	−0.014* (0.008)	0.050*** (0.014)	0.036* (0.019)	0.028*** (0.010)	−0.005 (0.008)
Post × Male	0.127*** (0.014)	0.077*** (0.012)	0.024 (0.019)	−0.018 (0.028)	0.019 (0.014)	−0.001 (0.012)
Constant	0.753*** (0.006)	0.892*** (0.005)	0.833*** (0.010)	0.840*** (0.014)	0.825*** (0.007)	0.930*** (0.006)
Observations	20,220	14,539	6,383	2,883	12,554	10,016
Mean dep. var.	0.659	0.845	0.813	0.838	0.814	0.902
R ²	0.048	0.020	0.022	0.006	0.007	0.011
F-stat.	336	101	47	5	30	39

Notes: This table shows the results of Difference-in-Differences regressions which capture the effect of the abolition of compulsory conscription on French citizenship acquisition. It is estimated on individuals born in France without French citizenship between 1974 and 1983, separately for 6 groups defined according to birth nationality (European Union, Other Europe, and Africa) and education (up to high school and above high school). European Union is defined as it was in 1996. The dependent variable, French citizenship acquisition, is a dummy variable taking the value 1 if the individual acquired French citizenship and 0 otherwise. The variable Male is a dummy variable taking the value 1 for males and 0 for females. The variable Post is a dummy variable taking the value 1 for individuals born after 1978, i.e., for cohorts not subject to compulsory conscription, and 0 otherwise. Hence, the interaction Post × Male is also a dummy variable taking the value 1 for males born after 1978 and 0 otherwise. Standard errors are reported in parentheses, and significance level is reported according to the following symbology. *p<0.1; **p<0.05; ***p<0.01. Source: *French Population Census*, wave 2014, INSEE.

Table B.4: Effect of the abolition of compulsory conscription on naturalization by education and birth nationality group - Robustness with European Union as it was before and after the 1995 European Union expansion

	French citizenship acquisition			
	European Union as in 1996		European Union as in 1994	
	Low edu.	High edu.	Low edu.	High edu.
Male	−0.222*** (0.008)	−0.128*** (0.008)	−0.222*** (0.008)	−0.127*** (0.008)
Post	0.022** (0.010)	−0.014* (0.008)	0.023** (0.010)	−0.012 (0.008)
Post × Male	0.127*** (0.014)	0.077*** (0.012)	0.126*** (0.014)	0.078*** (0.012)
Constant	0.753*** (0.006)	0.892*** (0.005)	0.753*** (0.006)	0.892*** (0.005)
Observations	20,220	14,539	20,198	14,484
Mean dep. var.	0.659	0.845	0.659	0.847
R ²	0.048	0.020	0.048	0.020
F-stat.	336	101	336	100

Notes: This table shows the results of Difference-in-Differences regressions which capture the effect of the abolition of compulsory conscription on French citizenship acquisition. It is estimated on individuals born in France with a non-French European Union citizenship between 1974 and 1983, for two definitions of the European Union, before and after the inclusion of Austria, Finland and Sweden in 1995, and separately for individuals whose education level is up to high school and individuals whose education level is above high school. The dependent variable, French citizenship acquisition, is a dummy variable taking the value 1 if the individual acquired French citizenship and 0 otherwise. The variable Male is a dummy variable taking the value 1 for males and 0 for females. The variable Post is a dummy variable taking the value 1 for individuals born after 1978, i.e., for cohorts not subject to compulsory conscription, and 0 otherwise. Hence, the interaction Post × Male is also a dummy variable taking the value 1 for males born after 1978 and 0 otherwise. Standard errors are reported in parentheses, and significance level is reported according to the following symbology. *p<0.1; **p<0.05; ***p<0.01. Source: *French Population Census*, wave 2014, INSEE.

Table B.5: Effect of the abolition of compulsory conscription on naturalization on European Union citizens

	French citizenship acquisition		
	(1)	(2)	(3)
Male	−0.209*** (0.006)	−0.209*** (0.006)	−0.187*** (0.006)
Post	0.015** (0.007)	0.050*** (0.009)	0.037*** (0.009)
Post×Male	0.115*** (0.009)	0.114*** (0.009)	0.119*** (0.009)
Age		0.007*** (0.001)	0.007*** (0.001)
Higher Education			0.161*** (0.005)
Constant	0.820*** (0.004)	0.559*** (0.043)	0.464*** (0.042)
Observations	34,759	34,759	34,759
Mean dep. var.	0.741	0.741	0.741
R ²	0.047	0.048	0.080
F-stat.	566	434	604

Notes: This table shows the results of Difference-in-Differences regressions which capture the effect of the abolition of compulsory conscription on French citizenship acquisition, progressively including controls for age and education. It is estimated on individuals born in France with a non-French European Union citizenship between 1974 and 1983. European Union is defined as it was in 1996. The dependent variable, French citizenship acquisition, is a dummy variable taking the value 1 if the individual acquired French citizenship and 0 otherwise. The variable Male is a dummy variable taking the value 1 for males and 0 for females. The variable Post is a dummy variable taking the value 1 for individuals born after 1978, i.e., for cohorts not subject to compulsory conscription, and 0 otherwise. Hence, the interaction Post × Male is also a dummy variable taking the value 1 for males born after 1978 and 0 otherwise. Standard errors are reported in parentheses, and significance level is reported according to the following symbology. *p<0.1; **p<0.05; ***p<0.01. Source: *French Population Census*, wave 2014, INSEE.

Table B.6: Effect of the abolition of compulsory conscription on citizenship acquisition

	Citizenship acquisition
ITT ($\hat{\tau}$)	0.107*** (0.015)
Group weights ($\hat{\omega}_g$)	
Female - European Union	0.065
Male - Other Europe	0.180
Female - Other Europe	0.334
Male - Africa	0.015
Female - Africa	0.158
Male - Arrived by age 6	0.140
Female - Arrived by age 6	0.108
Time weights ($\hat{\lambda}_t$)	
1992	0.000
1993	0.000
1994	0.000
1995	0.000
1996	1.000
Age	✓
Education	✓
Observations	106,081
Mean dep. var.	0.753

Notes: This table reports the results of a Synthetic Difference-in-Differences which estimates the effect of the abolition of compulsory conscription on naturalization rates among EU males. The first row displays the Intention-to-treat effect, estimated with the Synthetic Difference-in-Differences approach described in Section 5.1. Bootstrapped standard errors are reported in parentheses, and statistical significance is reported according to the following symbology. *p<0.1; **p<0.05; ***p<0.01. Group and time weights are computed following [Arkhangelsky et al. \(2021\)](#). Education and age are controlled for. Source: *French Population Census*, wave 2014, INSEE.

Table B.7: Effect of naturalization on public-sector jobs

	Public job
ITT ($\hat{\tau}$)	0.016** (0.007)
LATE ($\hat{\tau}/\hat{\gamma}$)	0.133** (0.063)
Group weights ($\hat{\omega}_g$)	
Female - European Union	0.166
Male - Other Europe	0.111
Female - Other Europe	0.153
Male - Africa	0.114
Female - Africa	0.188
Male - Arrived by age 6	0.120
Female - Arrived by age 6	0.148
Time weights ($\hat{\lambda}_t$)	
1992	0.000
1993	0.418
1994	0.582
1995	0.000
1996	0.000
Age	✓
Education	✓
Observations	106,081
Mean dep. var.	0.094
First-stage F-stat.	604
Share of compliers ($\hat{\gamma}$)	11.91%

Notes: This table reports the results of a Synthetic Difference-in-Differences which estimates the effect of the increase in the naturalization rates among EU males, induced by the abolition of compulsory conscription, on public-sector jobs. The first row displays the Intention-to-treat effect, estimated with the Synthetic Difference-in-Differences approach described in Section 5.1. The second row shows the Local Average Effect, computed as the ITT divided by the share of compliers. Bootstrapped standard errors are reported in parentheses, and statistical significance is reported according to the following symbology. *p<0.1; **p<0.05; ***p<0.01. Group and time weights are computed following [Arkhangelsky et al. \(2021\)](#). Education and age are controlled for. The F-statistic of the first stage and the share of compliers estimated in Appendix Table B.5 are reported at the bottom of the table. Source: *French Population Census*, wave 2014, INSEE.

Table B.8: Effect of naturalization on self-employment

	Self-employed
ITT ($\hat{\tau}$)	−0.011* (0.008)
LATE ($\hat{\tau}/\hat{\gamma}$)	−0.091* (0.065)
Group weights ($\hat{\omega}_g$)	
Female - European Union	0.055
Male - Other Europe	0.152
Female - Other Europe	0.159
Male - Africa	0.162
Female - Africa	0.149
Male - Arrived by age 6	0.158
Female - Arrived by age 6	0.165
Time weights ($\hat{\lambda}_t$)	
1992	0.305
1993	0.185
1994	0.001
1995	0.507
1996	0.001
Age	✓
Education	✓
Observations	106,081
Mean dep. var.	0.074
First-stage F-stat.	604
Share of compliers ($\hat{\gamma}$)	11.91%

Notes: This table reports the results of a Synthetic Difference-in-Differences which estimates the effect of the increase in the naturalization rates among EU males, induced by the abolition of compulsory conscription, on self-employment. The first row displays the Intention-to-treat effect, estimated with the Synthetic Difference-in-Differences approach described in Section 5.1. The second row shows the Local Average Effect, computed as the ITT divided by the share of compliers. Bootstrapped standard errors are reported in parentheses, and statistical significance is reported according to the following symbology. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Group and time weights are computed following Arkhangelsky et al. (2021). Education and age are controlled for. The F-statistic of the first stage and the share of compliers estimated in Appendix Table B.5 are reported at the bottom of the table. Source: *French Population Census*, wave 2014, INSEE.

Table B.9: Robustness of the ITT to the exclusion of the 1978 birth cohort

	With time weights		Without time weights	
	Including $t - 1$	Excluding $t - 1$	Including $t - 1$	Excluding $t - 1$
Employed	0.017** (0.009)	0.016** (0.009)	0.023*** (0.007)	0.021*** (0.008)
Unemployed	-0.006 (0.008)	-0.007 (0.008)	-0.007 (0.006)	-0.006 (0.007)
Inactive	-0.010* (0.006)	-0.009* (0.006)	-0.015*** (0.004)	-0.014*** (0.005)
Public job	0.016** (0.007)	0.016** (0.008)	0.016*** (0.005)	0.015*** (0.006)
Self-employed	-0.011* (0.008)	-0.011* (0.008)	-0.012** (0.006)	-0.012** (0.006)

Notes: This table reports the results of Synthetic Difference-in-Differences specifications which estimate the ITT effect of the increase in the naturalization rate of males born European Union citizens in France, induced by the abolition of compulsory conscription, on several labor market outcomes. The first two columns report the estimate for each outcome variable using the time weights as in [Arkhangelsky et al. \(2021\)](#), while the last two columns show the estimates resulting from an equal weighting of each birth cohort. The first column of each of these two sets includes all birth cohorts in the pre-period, while the second one is estimated without the 1978 birth cohort, i.e., the last one before the reform, which is potentially subject to anticipation effects. Bootstrapped standard errors are reported in parentheses, and statistical significance is reported according to the following symbology. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Source: *French Population Census*, wave 2014, INSEE.

Table B.10: Sensitivity of baseline results to the share of natives in the employment zone

	ITT		LATE	
	(1)	(2)	(3)	(4)
Employed	0.017** (0.009)	0.017** (0.009)	0.145** (0.079)	0.145** (0.078)
Unemployed	-0.006 (0.008)	-0.006 (0.008)	-0.054 (0.069)	-0.054 (0.067)
Inactive	-0.010* (0.006)	-0.010* (0.006)	-0.080* (0.052)	-0.080* (0.051)
Public job	0.016** (0.007)	0.016** (0.007)	0.133** (0.063)	0.134** (0.065)
Self-employed	-0.011* (0.008)	-0.011* (0.008)	-0.091* (0.065)	-0.091* (0.066)
Age	✓	✓	✓	✓
Education	✓	✓	✓	✓
%Natives in ZE		✓		✓

Notes: This table reports the results of the ITT effect and LATE of the increase in the naturalization rate of males born European Union citizens in France, induced by the abolition of compulsory conscription, on several labor market outcomes, with and without controlling for the share of natives in individuals' employment zone. ITTs are estimated with Synthetic Difference-in-Differences. LATEs are computed as the ITT divided by the share of compliers, estimated with Difference-in-Differences. Bootstrapped standard errors are reported in parentheses, and statistical significance is reported according to the following symbology. *p<0.1; **p<0.05; ***p<0.01. Source: *French Population Census*, wave 2014, INSEE.

Table B.11: Robustness of the LATE to the exclusion of the 1978 birth cohort

	With time weights		Without time weights	
	Including $t - 1$	Excluding $t - 1$	Including $t - 1$	Excluding $t - 1$
Employed	0.145** (0.079)	0.135** (0.075)	0.196*** (0.064)	0.180*** (0.068)
Unemployed	-0.054 (0.069)	-0.055 (0.068)	-0.055 (0.053)	-0.046 (0.061)
Inactive	-0.080* (0.052)	-0.072* (0.049)	-0.123*** (0.039)	-0.116*** (0.041)
Public job	0.133** (0.063)	0.135** (0.065)	0.138*** (0.047)	0.128*** (0.051)
Self-employed	-0.091* (0.065)	-0.091* (0.066)	-0.104** (0.053)	-0.099** (0.056)

Notes: This table reports the results of the local average treatment effect of the increase in the naturalization rate of males born European Union citizens in France, induced by the abolition of compulsory conscription, on several labor market outcomes. LATEs are computed as the ITT, estimated with Synthetic Difference-in-Differences, divided by the share of compliers, estimated with Difference-in-Differences. The first two columns report the estimate for each outcome variable using the time weights as in [Arkhangelsky et al. \(2021\)](#), while the last two columns show the estimates resulting from an equal weighting of each birth cohort. The first column of each of these two sets includes all birth cohorts in the pre-period, while the second one is estimated without the 1978 birth cohort, i.e., the last one before the reform, which is potentially subject to anticipation effects. Bootstrapped standard errors are reported in parentheses, and statistical significance is reported according to the following symbology. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Source: *French Population Census*, wave 2014, INSEE.